

**UNDERSTANDING THEMATIC STRUCTURE OF COMPUTER
SCIENCE RESEARCH FROM SCHOLARLY PUBLICATIONS OF
CENTRAL UNIVERSITIES**

MINOR PROJECT REPORT
SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENT FOR THE DEGREE OF

**MASTER OF SCIENCE
IN
COMPUTER SCIENCE**

SUBMITTED BY
RAJAT SHARMA (23234747050)

UNDER THE SUPERVISION OF
PROF. VIVEK KUMAR SINGH



**DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF MATHEMATICAL SCIENCES
UNIVERSITY OF DELHI
DELHI - 110007, INDIA**

ACADEMIC YEAR 2024-2025

Declaration

This is to certify that

- The work contained in the thesis is original and has been done by me under the supervision of my supervisor.
- The work has not been submitted to any other Institute for any degree or diploma.
- I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- Whenever I have used material (data, theoretical analysis, and text) from other sources. I have given due credit to them by citing them in the text of the thesis and giving their details in the references.

Date: December 2024

Place: Delhi

Rajat Sharma
(23234747050)

Certificate

This is to certify that this project work titled “Understanding Thematic Structure of Computer Science Research from Scholarly Publications of Central Universities”, submitted to the Department of Computer Science, University of Delhi, by Rajat Sharma (23234747050), has been carried out under my supervision. This work is done in partial fulfilment of the requirement for the completion of the Master of Science (M.Sc.) in Computer Science.

It is further Certified that this work is original and has been carried out under my guidance. To the best of my knowledge, this work has not been submitted for the award of any other title.

Date: December 2024

Place: Delhi

Prof. Vivek Kumar Singh

Supervisor

Head of Department

Department of Computer Science and
Dean, Faculty of Mathematical Sciences

University of Delhi

Delhi - 110007, India

Acknowledgement

I take this opportunity to extend my sincere thanks to all those who helped me in undertaking this project and devising the report.

I would like to extend my profound gratitude to my supervisor, Prof. Vivek Kumar Singh, Professor & Head, Department of Computer Science, and Dean, Faculty of Mathematical Sciences, University of Delhi, for his interest, guidance and valuable suggestions throughout this project. I feel honoured and privileged to work under his constant supervision. Without his advice, constructive ideas, positive attitude, and continuous encouragement, it would not have been possible to make such progress in the designated time frame. It has been a wonderful learning experience under his supervision and I hope to receive his guidance in future years as well.

I express my heartfelt gratitude to Dr. Prashasti Singh for supporting and guiding me towards my goal. Her invaluable guidance and support at every stage enabled me to understand and conceptualise the project both theoretically and practically. My research journey was profoundly shaped by her expertise, insightful feedback, and dedication. I also thank Dr. Abhirup Nandy for his suggestions, help and support provided during the project.

I also take this opportunity to thank my friends and contemporaries for their cooperation and compliance. I would also like to express my sincere appreciation to everyone whose support and contributions have been instrumental in the success of this endeavour, significantly enhancing my academic experience.

Last but not least, I am deeply grateful to the Almighty God for his everlasting blessings, and to my parents and well-wishers for their continuous love, good wishes, guidance, and encouragement, which have been invaluable throughout this journey.

Rajat Sharma
(23234747050)

Abstract

In the era of progressing advancements and rapidly evolving technologies it becomes important to investigate the major entities and their relevance in the contemporary scenario. The rapid development of computer science as a multidisciplinary topic has resulted in a tremendous expansion in its scholarly output in India. Scholarly data repositories like Web of Science (WoS), Scopus, Dimensions and so on provide a platform for data acquisition for subsequent analysis and research. In this work, the Web of Science scholarly database has been referred for obtaining the publication data. Initially, this study explores the research contributions of Indian central universities in computer science discipline from 2010 to 2023, followed by the primary objective of tracing the evolution of thematic trends in the domain of computer science by employing scientometric and text analytics methodologies. The computer science research article and review publications of the period of over twenty-five years ranging from 1999 through 2023 have been downloaded from the Web of Science database and analysed, leveraging burst detection, clustering and visualization techniques. The analysis highlights the dynamic character of computer science research, which shifts from theoretical frameworks to solving real-world problems and integrating cutting-edge mechanisms. This work gives a complete overview of the thematic evolution of computer science research in India, allowing for more informed decision-making and strategic advancements in academia and industry.

Keywords: Scholarly Databases, Computer Science, Research Output, CAGR, Burst Detection, Clustering.

Table of Contents

Chapter No.	Name	Page No.
	Declaration	i
	Certificate	ii
	Acknowledgement	iii
	Abstract	iv
	List of Tables	vii
	List of Figures	viii
	Abbreviations	ix-xi
Chapter 1	Introduction	1-6
	1.1 An overview of scholarly databases	1
	1.1.1 Web of Science	1
	1.1.2 Scopus	2
	1.1.3 Dimensions	2
	1.1.4 Google Scholar	2
	1.1.5 DBLP	3
	1.2 Scientometrics	3
	1.3 India's research scenario	4
	1.4 Thematic trajectories evolution analysis	5
	1.5 Organization of the report	6
Chapter 2	Related Work	7-9
	2.1 Studies on computer science multidisciplinary research	7
	2.2 Studies on scholarly databases	7
	2.3 Studies on analysis of research output from Indian Institutions	7
	2.4 Studies on analysing thematic research trends	8
Chapter 3	Data and Methodology	10-15
	3.1 Data acquisition and description	10
	3.1.1 For central universities' research output	10
	3.1.2 For the analysis of the thematic trajectories in computer science	12

	3.2 Methodology	13
	3.2.1 Data Pre-processing	14
	3.2.2 Burst Detection	14
	3.2.3 Clustering	15
	3.2.4 Word Cloud and Alluvial Diagram Representation	15
	3.2.4 Compound Annual Growth Rate	15
Chapter 4	Results	16-50
	4.1 Research output of central universities from 2010-2023	16
	4.2 Thematic trajectories analysis of research in computer science categories of Web of Science	21
	4.2.1 Burst detection results of individual periods	22
	4.2.2 Clustering results	35
	4.2.3 Results of graphical representation techniques	45
Chapter 5	Discussion and Conclusion	51-54
	5.1 Central universities' research output	51
	5.1.1 Compound annual growth rate (CAGR) of research output from Indian central universities	51
	5.2 Analysis of the thematic trajectories in computer science	52
	5.2.1 Integrative view of burst analysis, clustering and graphical representations	52
	5.3 Conclusion	53
	5.4 Future scope	54
	References	55-57
	Appendix	58-66

List of Tables

Table No.	Table Name	Page No.
1	Publications' count of different periods	12
2	Year Wise Research Output of the Central Universities in India in Computer Science Research during 2010-2023	18
3	Top 30 words in keywords analysed by burst detection algorithm (1999-2023)	23
4	Top 30 words in keywords analysed by burst detection algorithm (1999-2003)	25
5	Top 30 words in keywords analysed by burst detection algorithm (2004-2008)	27
6	Top 30 words in keywords analysed by burst detection algorithm (2009-2013)	29
7	Top 30 words in keywords analysed by burst detection algorithm (2014-2018)	31
8	Top 30 words in keywords analysed by burst detection algorithm (2019-2023)	33
9	Major Keywords in Different Periods	40
10	Inference from Alluvial Diagrams	47

List of Figures

Figure No.	Figure Name	Page No.
1	Methodology Framework Flowchart	13
2	Publication Count in Central Universities in Computer Science during 2010-2023	17
3	Volume of India's Computer Science Research Output during 1999-2023	21
4	Volume of India's Computer Science Research Output during the periods 1999-2003, 2004-2008, 2009-2013, 2014-2018 and 2019-2023 respectively	22
5	Top 30 keywords extracted by burst detection algorithm (1999-2023)	24
6	Top 30 keywords extracted by burst detection algorithm (1999-2003)	26
7	Top 30 keywords extracted by burst detection algorithm (2004-2008)	28
8	Top 30 keywords extracted by burst detection algorithm (2009-2013)	30
9	Top 30 keywords extracted by burst detection algorithm (2014-2018)	32
10	Top 30 keywords extracted by burst detection algorithm (2019-2023)	34
11	Clusters of Periods 1999-2003 and 2019-2023	40
12	Alluvial / Sankey Diagrams Representing the Evolution Keywords in Different Periods	47

Abbreviations

3D	Three Dimensional
AGRICOLA	Agricultural Online Access
AHCI	Arts and Humanities Citation Index
AI	Artificial Intelligence
AMU	Aligarh Muslim University
ANN	Artificial Neural Network
AR	Augmented Reality
AU	University of Allahabad
BERT	Bidirectional Encoder Representations from Transformers
BHU	Banaras Hindu University
BibTex	Bibliographic Text
BKCI	Book Citation Index
CAGR	Compound Annual Growth Rate
ChemRxiv	Chem-archive
CNN	Convolutional Neural Network
CoMFA	Comparative Molecular Field Analysis
CPCI	Conference Proceedings Citation Index
CSAB	Content Selection and Advisory Board
CSV	Comma Separated Values
CU	Country
DBLP	Database Logic Programming
DL	Deep Learning
DOAJ	Directory of Open Access Journals
DOI	Digital Object Identifier
DT	Document Type
ESCI	Emerging Sources Citation Index
I4OC	Initiative for Open Citations

IoT	Internet of Things
ISI	Institute of Scientific Information
IT	Information Technology
JMI	Jamia Millia Islamia
JNU	Jawaharlal Nehru University
JPD	Journal Packing Density
LSTM	Long Short Term Memory
MACD	Moving Average Convergence Divergence
ML	Machine Learning
NLP	Natural Language Processing
OI	ORCID Identifier
PT	Publication Type
PU	Pondicherry University
PY	Publication Year
QSAR	Quantitative structure–activity relationship
QSS	Quantitative Science Studies
RIS	Research Information Systems
RQ	Research Question
SCI	Science Citation Index
SSCI	Social Sciences Citation Index
TU	Tezpur University
TP	Total Publications
UGC	University Grants Commission
UoD	University of Delhi
VOS Viewer	Visualization of Science
VR	Virtual Reality
WC	Web of Science Category
WCSS	Within cluster sum of squares
WoS	Web of Science

WSN	Wireless Sensor Networks
XLS	Excel
XML	Extensible Markup Language

Chapter 1

Introduction

1.1 An overview of scholarly databases

Scholarly databases are specialised online platforms or digital repositories that offer access to academic and research-related resources such as journal articles, conference papers, theses, dissertations, books, and reports. These databases aim to help researchers, educators, and students by providing reputable, peer-reviewed, and established knowledge. Databases can generally be categorized into two types: subject-specific databases and multidisciplinary databases. Subject-specific databases focus on scientific output within particular disciplines, such as DBLP¹ (Computer Science), AGRICOLA² (Agriculture and related fields), and ChemRxiv³ (Chemistry and its subfields). On the other hand, multidisciplinary scholarly databases include scientific literature across a variety of fields in both science and the humanities. Well-known examples of multidisciplinary databases are Web of Science, Scopus, Dimensions, Google Scholar, OpenAlex etc. which are often used for information retrieval and research evaluation tasks. Among these databases, Web of Science is the oldest database, established in the year 1964, followed by Scopus and Google Scholar in the year 2004, Dimensions created in the year 2018, and OpenAlex being the most recent database, introduced in 2022. The special issue of Quantitative Science Studies⁴ provides a more in-depth analysis of various bibliographic data sources. Numerous scientometric studies, impact assessments, and research evaluation exercises are aided by the metadata of these scientific publications that are indexed by scholarly databases, and users can more easily access the research papers through links to the online publication texts.

1.1.1 Web of Science

Web of Science, the oldest scholarly database, was primarily developed as an information retrieval tool by Eugene Garfield from the Institute of Scientific Information (ISI). Originally known as the Science Citation Index (SCI), it initially covered 700 journals and was designed primarily as a citation index. Over time, it expanded to include additional citation indices: the Social Sciences Citation Index (SSCI), the Arts & Humanities Citation Index (AHCI), the Book Citation Index (BKCI), the Emerging Sources Citation Index (ESCI) and Conference Proceedings Citation Index (CPCI). If a journal is indexed by one of the citation indices, all of its articles are included in the Web of Science database. Presently, Web of Science is owned by Clarivate Analytics⁵.

¹ <https://dblp.org/>

² <https://agricola.nal.usda.gov/>

³ <https://chemrxiv.org/engage/chemrxiv/public-dashboard>

⁴ <https://direct.mit.edu/qss/issue/1/1>

⁵ <https://clarivate.com/webofsciencegroup/>

According to the most recent data⁶, the Web of Science Core Collection covers about 91 million scholarly publications and 2.19 billion cited references (tracing back to 1900) in 254 distinct subject domains. The Web of Science web interface offers various data search options like Basic Search, Advanced Search, Author Search etc., and download formats such as tab-delimited, CSV, xls (excel), RIS, plain text, BibTex etc. For basic usage, the downloaded data includes approximately 72 field tags, such as PT (Publication Type), DT (Document Type), OI (ORCID Identifier), and others. A thorough summary of the Web of Science database may be found in the research article by Birkle et al. (Birkle et al., 2019).

1.1.2 Scopus

Scopus database is a product of Elsevier, established in 2004. It is often regarded as one of the largest curated databases, encompassing scientific journals, books, conference proceedings, and more, all selected through a process of content selection followed by continuous re-evaluation. The decision to index a journal, conference, or other publication is made by the Content Selection and Advisory Board (CSAB). The Scopus platform offers data access through Search, Discover, and Analyse options. Search allows document, author, and advanced search, while Discover helps identify collaborators and research organizations. Analyse tracks citations and assesses search results. Data from Scopus includes 43 fields for research publications like source title, abstract, author keywords, document type etc. The research by Baas et al. (Baas et al., 2019), published in “Quantitative Science Studies” (QSS) in a special issue on academic databases, offers a thorough and detailed overview of the Scopus database.

1.1.3 Dimensions

The Dimensions database, established in 2018, is comparatively newer than Web of Science and Scopus. Dimensions offers a unified platform for accessing various types of research data, as highlighted in its launch description, which states, “it consolidates data that describes and links awarded grants, clinical trials, patents, policy documents, and altmetric information, alongside traditional publication and citation data” (Herzog et al., 2020). Unlike Web of Science and Scopus, Dimensions employs a distinct approach to data sourcing, with Crossref and PubMed serving as the “data spine”. The bottom-up approach means that data from Crossref and PubMed is further enriched by gathering information on affiliations, citations, and more. This enhancement process involves collecting data from multiple sources, including DOAJ, initiatives like Open Citations and I4OC, clinical trial registries, publicly available policy data, and other Digital Science companies such as Altmetric and IFI Claims. As the newest database, Dimensions draws from various sources, offering comprehensive data on institution IDs, grant IDs, and other related information. The studies by Thelwall, 2018; Martín-Martín et al., 2018a; Harzing, 2019; Visser et al., 2019 etc. provide thorough insights into the Dimensions database.

1.1.4 Google Scholar

⁶ <https://clarivate.com/webofsciencegroup/solutions/web-of-science-core-collection/>

Google Scholar, launched in 2004, is an academic search engine designed to help academicians and researchers find relevant work across the world of scholarly research⁷. It offers a single platform access to various disciplines and sources of scholarly literature, including peer-reviewed articles, theses, books, abstracts, letters, and editorials. Google Scholar's crawlers access databases of established scholarly publishers, university presses, digital hosts like MetaPress, Ingenta, HighWirePress, societies, government agencies, and reputable scholarly organizations, as well as digital reprint/preprint servers like arXiv.org, Astrophysics Data System, and CiteBase. Among the major crawler-based search engines, such as Google Scholar, Microsoft Academic, and Scirus, which made scholarly data freely accessible (Ortega, 2014), Google Scholar emerged as the leading source of information in academia (Van Noorden, 2014).

1.1.5 DBLP

DBLP Computer Science Bibliography is primarily an online database that catalogues publications in the field of Computer Science. It is designed to assist computer science researchers by offering free access to extensive, high-quality bibliographic metadata and links to online publication versions. Since 2018, Schloss Dagstuhl-Leibniz-Zentrum für Informatik (LZI)⁸, Germany, has been in charge of conducting the operation and maintenance of DBLP. The DBLP user interface offers search functionality through journals, authors/editors, conferences/workshops, monographs, and series. The DBLP data is available for download as an XML file. An extensive explanation of the DBLP records and their properties may be found in the study carried out by Ley (Ley, 2009).

In conclusion, numerous scholarly databases are available globally. These databases play a crucial role in advancing research by making scholarly resources readily accessible to researchers, academics, and professionals worldwide. Some of these platforms operate on a subscription-based model, ensuring comprehensive and high-quality content, while others promote open access, fostering inclusivity and wider dissemination of scientific knowledge. This diversity in access models contributes to the broad reach and impact of academic research across various fields and disciplines.

In this work, we have utilised the Web of Science scholarly database to extract data for conducting analyses reported under different chapters. Web of Science was selected as it is one of the leading databases commonly used for conducting a variety of scientometric studies.

1.2 Scientometrics

Scientometrics is a methodology for appraising the “quality” of science and analyzing advancements in science, technology, and innovation. It focuses on measuring the impact of scientific publications and is often conducted using bibliometric techniques. It covers all quantitative aspects and characteristics of the study of science, scientific communication,

⁷ <https://scholar.google.com/intl/en/scholar/about.html>

⁸ <https://www.dagstuhl.de/en/>

and science policy. It involves the analysis, assessment, and graphical representation of information related to science and technology. Scientometrics refers to the communication process in science, including social and cultural components, and is often used interchangeably with “science of science”. Scientometrics is a discipline that uses quantitative approaches to analyze science as an information process.

In the realm of scholarly publication data, Scientometrics refers to the systematic measurement of scientific output, particularly through the analysis of publications and their citations.

The growth of science and data has led to an increase in the volume of publications and research outputs, while advancements in computers and technology have enabled more effective data analysis. Whereas the social impact of science examines the behaviour of scientists, Scientometrics centres on the analysis of scientific publications.

Scientometrics, a term originating from Russia, pertains to the application of quantitative methods to the study of the history of science. Derek de Solla Price, a key figure in the field, introduced the concept of examining documents and articles through networks in his seminal 1965 paper, “Networks of Scientific Papers”. The study of networks has since become a fundamental aspect of Scientometrics. Furthermore, the academic journal “Scientometrics” was established in 1978, further advancing the field.

As the evaluation and ranking of scientists and institutions gained greater attention, the importance of scientific paper networks became increasingly evident. Scientometrics, which involves the bibliometric analysis of publications and citations, has become essential in understanding the impact of research and identifying evolving trends across scientific domains. One notable example is the Academic Ranking of World Universities, first published in 2004 by Shanghai Jiao Tong University, which uses bibliometric data to assess and rank global institutions. In addition to institutional evaluations, scientometrics is also employed to investigate and analyze the thematic trajectories of various research domains. This method provides valuable insights into the development and evolution of specific fields, such as computer science, through the study of scholarly metadata from databases like the Web of Science. Various tools and software applications, such as VOS Viewer⁹, CiteSpace¹⁰ and Sci2¹¹, were developed to facilitate specific analyses and visualizations. Cobo et al. compare and review various software tools for science mapping, focusing on their bibliometric techniques and analysis methods to understand the conceptual, intellectual, and social structure of specific disciplines (Cobo et al., 2011).

1.3 India’s research scenario

⁹ <https://www.vosviewer.com/> [VOSViewer: Visualizing Scientific Landscapes]

¹⁰ <https://citespace.podia.com/>

¹¹ <https://sci2.cns.iu.edu/user/index.php> [Sci2 Team(2009), Science of Science Tool, Indiana University and SciTech Strategies]

An educated populace with the applicable knowledge, skills, and attitudes is crucial for fostering economic and social progress in the twenty-first century. Similarly, in the current era of knowledge-based economies, research and innovation are critical to national growth. Developed countries recognized the value of higher education, research, and innovation and took proactive measures to boost this sector. Universities are globally recognized as hubs for education, research, and innovation. Developing a strong higher education and research infrastructure is crucial for a country's success. Information (as at the time of this study) from the University Grants Commission (UGC)¹², New Delhi, suggests that there are around 1189 universities in the country. Currently, this system consists of 57 central, 494 state, 138 deemed and 501 private universities and more than 150 institutes of National Importance. Consequently, it becomes imperative to examine the research contributions of universities and to explore the research landscape by identifying the most prominent themes across various disciplines and categories. A number of prior studies have examined the research output of different groups of institutions (Marisha et al., 2017; Basu et al., 2016; Prathap, 2014; Kanaujia et al., 2022).

1.4 Thematic trajectories evolution analysis

Thematic evolution analysis in scientific research reveals important trends, movements, and growing areas of interest within a specific topic across time. Numerous studies have been conducted to examine the trends in research across different domains (Lopez-Lopez et al., 2024; Hoonlor et al., 2013; Sohrabi et al., 2019; Singh & Kumari, 2021). Scholarly databases like Web of Science and Scopus were referred for undertaking such studies. Web of Science has around 254 (as of the time the current study was conducted) subject categories¹³ in total spread across a multitude of disciplines. In Computer Science there are seven subject categories: Computer Science, Artificial Intelligence; Computer Science, Cybernetics; Computer Science, Hardware & Architecture; Computer Science, Information Systems; Computer Science, Interdisciplinary Applications; Computer Science, Software Engineering; Computer Science, Theory & Methods. In this work, we have taken into consideration these subject categories of Computer Science for the analysis.

In some studies like Tattershall et al., 2020 and Kleinberg, 2002, techniques like MACD (Moving Average Convergence Divergence) and Kleinberg's burst detection algorithm have been discussed for exploring the trending terms across various domains like computer science. Detecting bursty phrases in scientific literature could have numerous applications like early detection can help funding organizations and publishers identify promising new concepts and focus their support accordingly. For newcomers to a field and researchers studying the sociology of science, an automated list of trending topics over time would provide a quick overview of the field's evolution. Additionally, compiling a list of historical bursty terms could facilitate the analysis of the life cycles that new ideas undergo as they mature and evolve.

¹² <https://www.ugc.gov.in/universitydetails/university?type=MUOh4z0uqRaY2k8Ag10I0g==>

¹³ <https://webofscience.zendesk.com/hc/en-us/articles/27505726032017-Web-of-Science-Subject-Categories>

In this study, initially, we analysed the research output of the 46 Central Universities¹⁴ of India from 2010 through 2023 and computed the CAGR (Compound Annual Growth Rate) of the Central Universities. Further, we explored the evolution of research themes in computer science in India from 1999 through 2023. For this, we employed a two-step analytical strategy, first burst detection analysis to identify moments of strong emphasis and abrupt expansion within distinct themes, and then clustering analysis to aggregate and depict thematic trajectories. Additionally, we used visual tools like word clouds and alluvial diagrams, which provided more insights into the importance of themes and their interrelationships. This combination of methods allowed us to gain deeper insights into conceptual developments throughout the domain and new issues over the previous two decades. By utilizing these analytical approaches, the study seeks to answer key research questions related to the academic performance of the central universities and the evolution of research themes in the field, thus contributing to a deeper understanding of the subject matter. More specifically the study attempts to answer the following research questions.

RQ1. What is the quantum of computer science research in India over a period of time?

RQ2. What is the growth rate of central universities of India in terms of computer science research output?

RQ3. What are the different ways to map the evolution of thematic trajectories in Computer Science research for India?

RQ4. What are the historical and current bursts in computer science literature?

To address these questions, the study examines and analyzes research publication data on Computer Science in India using Scientometric analysis. The Scientometric analysis involves the quantitative study of science, technology, and innovation, applied to explore extensive scientific literature and provide a comprehensive review (Leydesdorff & Milojević, 2012). Additionally, text analytics techniques have been employed to extract deeper insights into research growth and the thematic structure of the field. The study's findings add to a better knowledge of the state of computer science research in India, providing useful information for researchers, policymakers, and industry stakeholders.

1.5 Organization of the report

The rest of the report is organized as follows: In Chapter 2 an in-depth analysis of relevant research papers that have informed and influenced the work conducted in this project has been provided. Chapter 3 outlines the details of the data and methodology used to perform the research analysis. In Chapter 4, we present the results and analysis of the experiments, highlighting the key findings of our work. In Chapter 5, we discuss the inferences obtained from the work and put forward our conclusions and potential directions for future research related to the study.

¹⁴ <https://cup.ugc.ac.in/Home/CUList>

Chapter 2

Related Work

This chapter provides a summary of earlier studies and available literature that serve as the foundation for the current research. It investigates major contributions to the subject, identifies knowledge gaps, and emphasizes approaches and frameworks pertinent to the research. This chapter sets the stage for the study by critically analyzing relevant research and illustrating how it connects with and expands on earlier discoveries.

2.1 Studies on computer science multidisciplinary research

The evolution of computer science as a field has been comprehensive. It has had an impact on practically every element of human life, enabling remarkable advances in communication, healthcare, engineering, sustainability, and other fields. The current research scenario demonstrates its application in many other disciplines such as biomedical sciences, mathematics, physical sciences, simulations, and so on. Many scholars throughout the world have used computer science approaches to carry out their studies (Shortliffe & Blois, 2006; Shortliffe et al., 2014; Rangayyan & Krishnan, 2024; Li et al., 2024).

2.2 Studies on scholarly databases

With a prolific amount of scholarly data being accumulated, it establishes avenues for advanced research on them. There are various subject-specific and multidisciplinary scholarly databases like Web of Science, Scopus, Google Scholar etc. (Heck et al., 2024). While many studies have explored the coverage of scholarly databases (Singh et al., 2021; Gusenbauer, 2022; Mongeon & Paul-Hus, 2016; Michels & Schmoch, 2012) others have explored the citation counts in major scholarly databases (Singh et al., 2022b; Visser et al., 2021; Martín-Martín et al., 2021; Rovira et al., 2019; Martín-Martín et al., 2018b; Mingers & Lipitakis, 2010).

There have also been studies that have focused on investigating the relationship between a country's journal indexing and its research output in scholarly databases (e.g., Basu, 2010; Leta, 2011; Michels & Schmoch, 2012; Najari & Yousefvand, 2013; Collazo-Reyes, 2014; Bhattacharya et al., 2015; Erfanmanesh et al., 2017; Moed et al., 2018 etc.).

2.3 Studies on analysis of research output from Indian Institutions

Several previous studies have assessed the research output of different universities and institution systems in India (Marisha et al., 2017; Basu et al., 2016; Prathap, 2014; Singh et al., 2022a). Marisha et al. 2017, examined the research performance of 39 Central Universities in India during a 15-year period (1990–2014). The study assessed the productivity, productivity per capita, growth rate of research output, authorship and cooperation patterns, citation impact, and discipline-specific research strengths of those

universities. The results highlight each university's research capabilities in specific disciplines.

Basu et al., 2016 in their research, offered a multidimensional 'Quality-Quantity' Composite Index for bibliometric data used to rank Central Universities in India. The measure is comparable to the h-index, as well as the size-dependent Leiden ranking and the University Ranking by Academic Performance. A generalized model for the index based on other variables and variable weights is proposed. This method can be used to rank and make decisions at the national or regional levels.

The work carried out by Prathap, 2014 evaluated the performance of select higher education institutions in India's academic research output from 2003-2011 using datasets from the 2013 SCImago Institutions Rankings World Reports. All institutions in this elite category show reasonable to high growth rates in output, but high performers, mainly national-important institutions, are at a relatively low growth level. Some institutions also show promisingly high rates of improvement in research quality.

Singh et al., 2022a established a new scientometric portal called Indian Science Reports, which is available at www.indianscience.net¹⁵. Indian Science Reports is an online portal that serves as a single, integrated resource for analytical data about India's research competencies and institutions. It gauges India's research performance, such as output, citations, highly cited publications, international collaboration, open access levels, gender distribution, and social media presence. The portal also allows users to look for the research performance of key Indian higher education institutions and identify the best-performing colleges in certain research areas.

2.4 Studies on analysing thematic research trends

In modern times, with computer science developing as one of the most versatile and extensively applicable disciplines, frequently interacting with other fields, there has been a greater emphasis on multidisciplinary studies. This expanding integration has spurred an interest in studying trends and the growth of thematic trajectories in computer science (Hoonlor et al., 2013; Tattershall et al., 2020; Singh & Kumari, 2021; Mahmoud, 2024).

The significance of big data analysis in modern research, the significance of machine learning and artificial intelligence in data science, and the interdisciplinary applications of big data were explored in a work by Mahmoud, 2024. The editorial highlights the growing reliance on AI and ML to detect patterns in data that are often beyond human capabilities. It also emphasizes the interdisciplinary aspect of data science, with applications in biology, chemistry, physics, and engineering.

Jurayeva & No, 2024 investigated the incorporation of IT into medical and biomedical engineering education in 2024, as education undergoes a digital change. Their findings show

¹⁵ www.indianscience.net

that combining IT skills provides students with critical competencies for current healthcare contexts, encouraging creativity and improving patient outcomes.

In the recent era, the computer science discipline has gained broader attention with significant advances in data collection and aggregation, analytics and availability of suitable computer processing power and techniques. This makes it important to determine the evolution of the research trends in the field of Computer Science. Numerous studies have been conducted to examine the trends in research across different domains of computer science (Lopez-Lopez et al., 2024; Hoonlor et al., 2013; Sohrabi et al., 2019; Singh & Kumari, 2021).

In the work by Kostelić & Etinger, 2024 a bibliometric investigation of scientific efforts addressing cybersecurity concerns in the metaverse is provided. They used bibliometric tools to map significant research trajectories, highlight common cybersecurity problems, and examine emerging trends in the area. Their findings emphasize the critical importance of strong security measures in virtual settings, as well as the complexities of resolving these challenges.

The research article by Singh et al., 2023, investigated the evolution of concept trajectories in AI for Sustainable Development Goals. They employ bibliometrics, path analysis and content analysis techniques to discover the trends of AI in SDG-related research.

Tattershall et al., 2020 performed a study to determine bursty terms in the field of computer science, by using MACD (Moving Average Convergence Divergence) a trend detection algorithm from stock market analysis to analyze over 30 years of computer science research abstracts. The term “prevalence” is treated like a stock price, and the free text of abstracts and titles is used for a finer-grained analysis. The authors report a list of bursty terms and use historical data to build a classifier to predict their future popularity, achieving an accuracy of 80%.

The work by Kleinberg, 2002 studied the phenomenon of burst-a rapid and persistent increase in population in comparison to a historical baseline level, and proposed an algorithm called Kleinberg's burst detection algorithm. It demonstrates extensive attempts to analyze research trends, theme advances, and productivity across multiple domains.

However, there are still gaps in understanding specific features of research evolution in targeted contexts. This chapter establishes the framework for the current investigation by outlining the gaps and emphasizing the importance of a full analysis. The incorporation of the concepts of clustering along with visual and diagrammatic representations like alluvial diagrams is a unique approach that offers a deeper understanding of the topic. Building on previous research, this study intends to contribute to a better understanding of research patterns and theme progressions in computer science research in India over the last 25 years, addressing identified gaps in the existing body of information.

Chapter 3

Data and Methodology

3.1 Data acquisition and description

Two kinds of data have been utilised in the analysis, one for Central Universities' Research Output and another for the Analysis of the Thematic Trajectories in Computer Science.

3.1.1 For central universities' research output

For the research output data for central universities is obtained from the Web of Science scholarly database, for the period 2010 to 2023. To get the data from Web of Science, an advance search query Advanced search query: “(((WC=(Computer Science, Theory & Methods OR Computer Science, Software Engineering OR Computer Science, Interdisciplinary Applications OR Computer Science, Information Systems OR Computer Science, Hardware & Architecture OR Computer Science, Cybernetics OR Computer Science, Artificial Intelligence)) AND DT=(Article OR Review)) AND PY=(2010-2023)) AND OG=(Aligarh Muslim University OR Assam University OR Babasaheb Bhimrao Ambedkar University OR Banaras Hindu University (BHU) OR Central University of Gujarat OR Central University of Haryana OR Central University of Himachal Pradesh OR Central University of Jammu OR Central University of Jharkhand OR Central University of Karnataka OR Central University of Kashmir OR Central University of Kerala OR Central University of Punjab OR Central University of Rajasthan (CURAJ) OR Central University of South Bihar OR Central University of Tamil Nadu OR Dr. Hari Singh Gour University OR Guru Ghasidas Vishwavidyalaya OR University of Delhi OR Hemwati Nandan Bahuguna Garhwal University OR Jamia Millia Islamia OR Jawaharlal Nehru University, New Delhi OR Manipur University OR Mizoram University OR Nagaland University OR North Eastern Hill University OR Pondicherry University OR Rajiv Gandhi University, Itanagar OR Sikkim University OR Tezpur University OR Indira Gandhi National Tribal University OR Tripura University OR University of Allahabad OR University of Hyderabad OR Visva Bharati University)”, was used to get the publication records of the central universities which were available in the Web of Science search. This query fetched 3,647 documents and we obtained full publication records of the data.

The list of central universities¹⁶ referred to in the study is as follows,

- Aligarh Muslim University
- Assam University
- Babasaheb Bhimrao Ambedkar University
- Banaras Hindu University
- Central Sanskrit University
- Central Tribal University of Andhra Pradesh
- Central University of Andhra Pradesh

¹⁶ <https://cup.ugc.ac.in/Home/CUList> [referred on September 25, 2024]

- Central University of Gujarat
- Central University of Haryana
- Central University of Himachal Pradesh
- Central University of Jammu
- Central University of Jharkhand
- Central University of Karnataka
- Central University of Kashmir
- Central University of Kerala
- Central University of Odisha
- Central University of Punjab
- Central University of Rajasthan
- Central University of South Bihar
- Central University of Tamil Nadu
- Dr. Harisingh Gour Vishwavidyalaya
- Guru Ghasidas Vishwavidyalaya
- Hemvati Nandan Bahuguna Garhwal University
- Jamia Millia Islamia
- Jawaharlal Nehru University
- Mahatma Gandhi Antarrashtriya Hindi Vishwavidyalaya
- Mahatma Gandhi Central University
- Manipur University
- Maulana Azad National Urdu University
- Mizoram University
- Nagaland University
- National Sanskrit University
- North Eastern Hill University
- Pondicherry University
- Rajiv Gandhi University
- Shri Lal Bahadur Shastri National Sanskrit University
- Sikkim University
- Sindhu Central University
- Tezpur University
- The English and Foreign Languages University
- The Indira Gandhi National Tribal University
- Tripura University
- University of Allahabad
- University of Delhi
- University of Hyderabad
- Visva Bharati

List of Universities' names absent on Web of Science Affiliations list:

- Central University of Sanskrit
- Central Tribal University of Andhra Pradesh
- Central University of Andhra Pradesh
- Central University of Odisha
- Mahatma Gandhi Antarrashtriya Hindi Vishwavidyalaya

- Maulana Azad National Urdu University
- National Sanskrit University
- Shri Lal Bahadur Shastri National Sanskrit University
- Sindhu Central University
- The English and Foreign Languages University

Therefore the study was performed for the universities present in the affiliations list of the Web of Science (WoS).

3.1.2 For the Analysis of the Thematic Trajectories in Computer Science

Another dataset from the scholarly database of WoS was utilised for the period of 1999-2023 in the five-year time frames of 1999-2003, 2004-2008, 2009-2013, 2014-2018, and 2019-2023 for the analysis of the evolution of thematic trends in computer science. Advance searches were conducted to obtain the metadata of the respective time frames. The query formulated was of the form:

((PY=(X-Y)) AND DT=(Article OR Review)) AND WC=(Computer Science, Theory & Methods OR Computer Science, Software Engineering OR Computer Science, Interdisciplinary Applications OR Computer Science, Information Systems OR Computer Science, Hardware & Architecture OR Computer Science, Cybernetics OR Computer Science, Artificial Intelligence)) AND CU=(India)

where, X-Y is the different range of years namely; 1999-2003, 2004-2008, 2009-2013, 2014-2018 and 2019-2023.

The **Table 1** contains the count of the documents fetched in the different year ranges and the number of pre-processed documents retrieved out of them.

Table 1: Publications' count of different periods

S.No.	Years	Total Documents Fetched	PreProcessed Data (dropped duplicated DOI)
1	1999-2003	1,986	1,441
2	2004-2008	3,715	2,612
3	2009-2013	5,378	5,095
4	2014-2018	13,406	12,947
5	2019-2023	44,756	44,403

3.2 Methodology

The methodology employed for the work is designed to systematically address the research questions pertaining to the growth, volume, and thematic evolution of computer science research in India. To analyze the growth rate and research output from central universities, a comprehensive dataset of research publications was curated. The study further delves into the quantum of computer science research over an extended period, identifying trends and patterns that illustrate its trajectory. Additionally, thematic evolution trajectories were explored using advanced analytical techniques, enabling the identification of key themes and their progression over time. This multifaceted methodological approach ensures a robust and comprehensive examination of the research landscape in computer science.

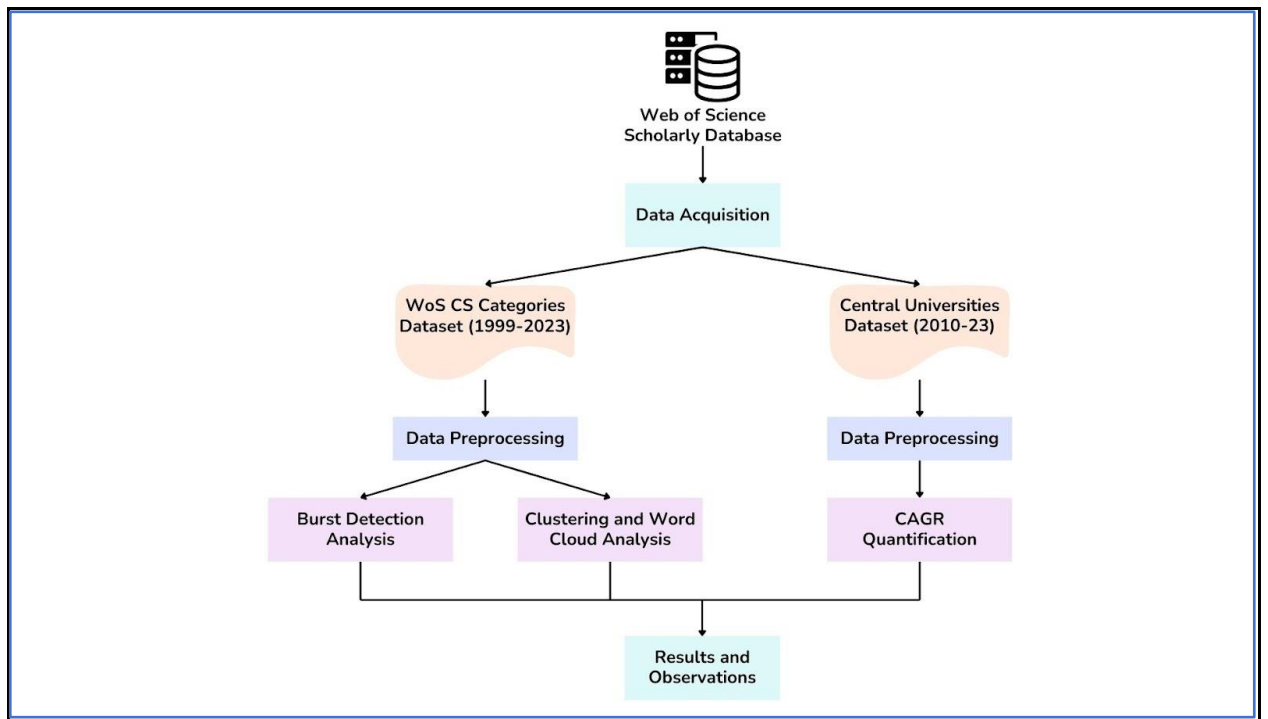


Figure 1: Methodology Framework Flowchart

Techniques like burst analysis (Kleinberg, 2002), clustering, word cloud, alluvial diagram, CAGR etc have been employed in this study to perform the research and analysis. **Figure 1** depicts the methodology framework of the proposed work. The datasets from Web of Science (WoS) have been extracted for the analysis.

The first dataset consists of metadata of the publications in the Research Articles and Review Papers of the Indian Central Universities in WoS seven Computer Science categories mentioned above in the data description section from the year 2010 through 2023. The data is preprocessed to remove the duplicated records and make data ready for subsequent analysis. Thereafter, graphical and visual representations were employed to gain a deeper

understanding of the research outputs from central universities, after which the Compound Annual Growth Rate (CAGR) was determined.

The second dataset is also gathered from the WoS database. It consists of India's publications categorized as Articles and Reviews across the various time periods analyzed in the study. The "Author Keywords" and "Publication Year" fields are used for the analysis. This data is utilised to appraise evolving bursty terminologies in computer science in specified periods of time. Further, the records are integrated to collectively analyse the data of 1999-2023 and results have been observed. The steps involved in the study are data preprocessing, burst detection, clustering, word cloud generation, and alluvial diagram representation. These steps are described as follows:

3.2.1 Data Pre-processing: After data collection, pre-processing is performed. The data is processed to eliminate irrelevant and unwanted entities from the original dataset ensuring more accurate results.

1. Cleaning involves eliminating redundant and irrelevant terms that do not contribute to the analysis or are unrelated to the research topic, such as ©-Elsevier-Paper, All rights reserved, Copyright, Published by Ltd-Inc., ASIS&T, The Authors-LLC, John Wiley & Sons, and similar phrases.
2. Filtering involves removing stop words that are irrelevant to the analysis, as well as discarding tokens that have fewer than three characters using suitable filters.
3. Tokenization is the process of splitting scientific texts into individual units, known as tokens, to facilitate the application of algorithms and the extraction of analysis.
4. Stemming involves identifying and merging different word suffixes and prefixes into their base forms to shorten words, such as converting "stems," "stemmer," "stemming," and "stemmed" into the root form "stem."

The combination of burst detection and clustering enhances the overall analytical process. Burst detection helps identify how computer science has developed over the past decades and highlights the key research areas up until the conclusion of the analysis (2023). Essentially, burst detection reveals the primary research topics that have been the focus of attention over the analysis period and offers insight into the progression of research. In parallel, clustering analysis provides a deeper understanding of the current research landscape. It elaborates on the key research areas identified by burst detection, offering a detailed analysis of those clusters based on the major themes highlighted by the burst detection results.

3.2.2 Burst Detection: Burst detection is a technique used in Scientometrics to detect substantial increases in the frequency of certain phrases or study subjects over time. It reveals developing trends, shifts in academic concentration, or breakthroughs by emphasizing periods of increasing activity, such as a surge in publications, citations, or conversations. This approach is useful for following the evolution of scientific knowledge and identifying areas of increased interest or innovation. The idea of burst detection was primarily proposed by Kleinberg, in 2002 (Kleinberg, 2002) through an algorithm which is known as

Kleinberg's burst detection algorithm. Sci2¹⁷ application software has been used for burst detection analysis.

3.2.3 Clustering: Clustering is a data analysis approach that puts related things together based on shared properties. In research, it is used to divide publications or subjects into discrete groups, so revealing patterns and trends within an area. In this work, the VOS Viewer¹⁸ application and K-mean algorithm are used for clustering the author keywords over the publication years. To determine the optimal number of clusters (k) in the algorithm elbow method has been incorporated. The Elbow Method is a strategy for calculating the ideal number of clusters (k) in K-means clustering. It operates by graphically plotting the within-cluster sum of squares (WCSS) or inertia for various values of k and determining the "elbow" point at which the rate of reduction in inertia slows or is minimal.

3.2.4 Word Cloud and Alluvial Diagram Representation: A word cloud is an image representation of text, with the size of each word proportional to its frequency in the dataset. Common words display bigger, allowing us to rapidly recognize major themes or concepts. Word clouds are used in text analysis, social media, and surveys to highlight relevant subjects or keywords. They are also called tag clouds. An alluvial diagram is a flow visualization that highlights relationships and changes over time or between groups, illustrating how data moves from one category to another. It is excellent for demonstrating patterns in multidimensional data. Author Keywords through time have been used for plotting the word clouds and alluvial diagrams to make a deeper understanding of the evolving thematic trends in computer science research.

3.2.4 Compounded Annual Growth Rate Computation: The compounded annual growth rate, or CAGR, is a metric that calculates the average annual growth rate of a business, investment, or research output over a given time period, assuming that the growth occurs at a constant pace. It is frequently used to compare the growth of various variables over time or to evaluate the growth of financial investments. In our work, we have employed it to examine the research progress of some central universities in computer science from 2010-2023. Following is the formula used for obtaining CAGR (Compound Annual Growth Rate):

$$CAGR(t_0, t_n) = \left(\frac{V(t_n)}{V(t_0)} \right)^{\frac{1}{t}} - 1$$

Where; $V(t_0)$ is the initial value, $V(t_n)$ is the end value, t_0 is the initial year, t_n is the final year, and t is the number of years, i.e., $(t_n - t_0)$.

Building on the methodological approaches outlined in this chapter, the next chapter presents the results derived from this study. The analysis explores the research output and thematic evolution within the domain of Computer Science, providing valuable insights into the growth and trends observed over the specified period.

¹⁷ <https://sci2.cns.iu.edu/user/index.php>

¹⁸ <https://www.vosviewer.com/>

Chapter 4

Results

The main conclusions and observations drawn from the analysis carried out in this study are presented in this chapter. In order to answer the previously stated research questions, the findings are methodically arranged to offer a thorough grasp of research trends, thematic evolution, and growth patterns. The primary goal is to provide clear and detailed insights into the data to support the study's objectives.

The proposed methodology is utilized to interpret and appraise the research horizons of computer science research in India. The results have been organised into two major parts further divided into sub-parts. *Firstly*, the research output volume of Central Universities from 2010-2023 is presented, followed by determination of their CAGR. *Secondly*, the thematic analysis of research in Computer Science categories of Web of Science from 1999-2023 has been shown.

4.1 Research output of central universities from 2010-2023

A dataset of 3,648 total publications (TP) was constituted from the WoS database for the years 2010 to 2023. From **Figure 2**, it can be seen that during 2010 to 2023, the number of publications in computer science at central universities have increased steadily. Publications in the early years, such as 2010–2013, were small, making up less than 3% of the total. However, a notable growth started in 2014 and continued to rise sharply from 2020 onward, with publication counts rising continuously above 10% of the total, reaching a peak of 16.146% in 2022. This suggests that computer science research output is becoming increasingly important, probably as a result of technological breakthroughs and more financing for programs for study. The collective CAGR, of all the central universities considered in this study, was found to be 18.35% which indicates a positive growth in the research output of these central universities.

The Compounded Annual Growth Rate (CAGR) in research publications from central universities from 2010 to 2023 is highlighted in **Table 2**, which shows notable growth differences. Exceptional increases in research output are shown by high-growth institutions like Indira Gandhi National Tribal University (45.65%), Central University of Haryana (71.88%), and Central University of Himachal Pradesh (73.21%). These increases are probably the result of concentrated efforts on research development in more recent universities. Prominent establishments such as Banaras Hindu University (13.18%), the University of Delhi (13.47%), and Jamia Millia Islamia (28.86%) show steady and consistent growth, indicating ongoing research productivity.

Rapid progress is demonstrated by up-and-coming contributors like Babasaheb Bhimrao Ambedkar University (25.99%) and Sikkim University (25.85%), suggesting a growing focus on research in formerly underrepresented regions. Nonetheless, some universities show negative or little growth, like the Central University of Gujarat (-23.53%) and the

Central University of Kerala (-9.43%). This could be due to a lack of research projects or infrastructure issues. Although there is considerable variation amongst institutions, the data generally shows an increase in publication output, highlighting the necessity of focused assistance to improve research performance in underperforming universities.

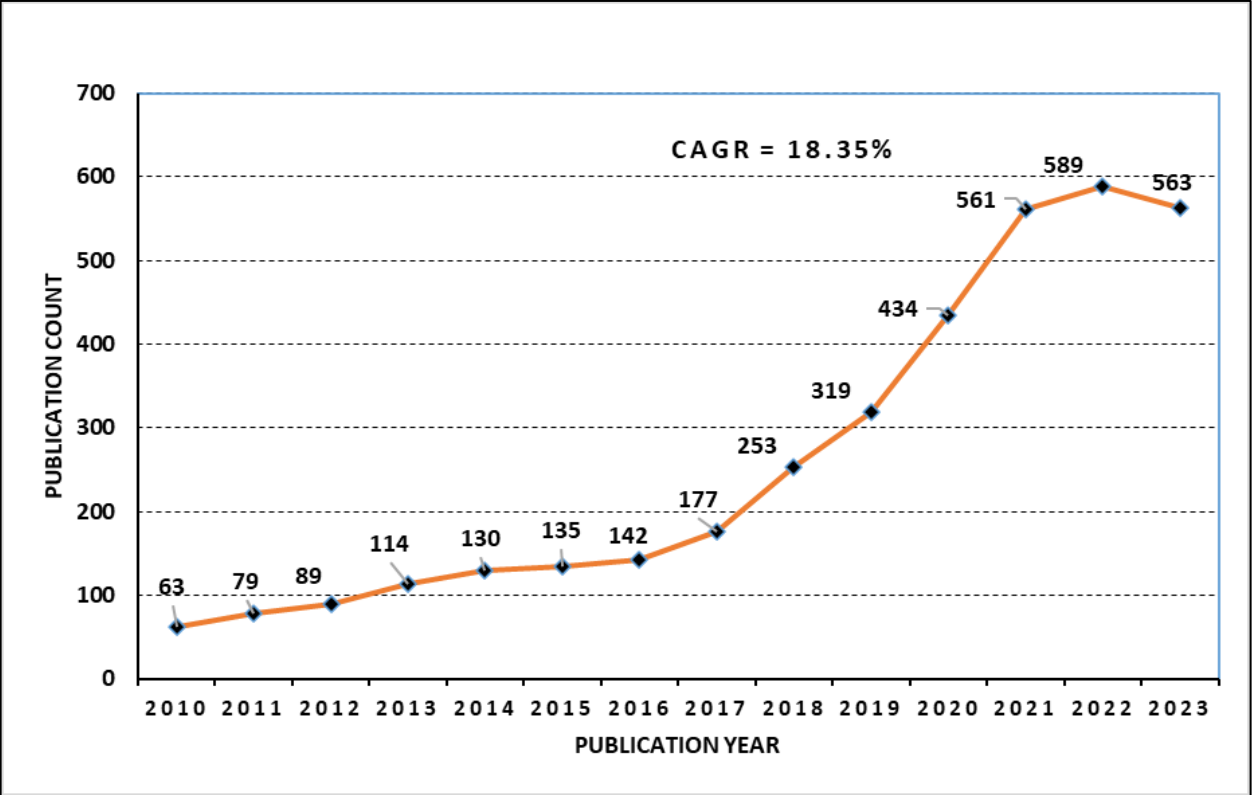


Figure 2: Publication Count in Central Universities in Computer Science during 2010-2023

As depicted in **Table 2**, some established universities like the University of Delhi (UoD), Banaras Hindu University (BHU), Aligarh Muslim University (AMU), Jawaharlal Nehru University (JNU), University of Allahabad (AU), Jamia Millia Islamia (JMI), Pondicherry University (PU), and Tezpur University (TU) demonstrate consistently greater number of publications as compared to other universities.

Table 2: Year Wise Research Output of the Central Universities in India in Computer Science Research during 2010-2023

S.No.	Institute	TP (2010- 2023)	Year														CAGR (in %)
			2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	
1	University of Delhi	494	12	14	10	16	28	20	20	27	48	44	45	71	77	62	13.47
2	Jawaharlal Nehru University	428	5	7	11	7	24	18	30	21	37	51	55	58	53	51	19.56
3	Banaras Hindu University (BHU)	398	13	9	9	16	6	13	17	18	31	34	57	42	68	65	13.18
4	Aligarh Muslim University	345	12	8	9	8	5	7	9	14	15	31	32	77	60	58	12.88
5	Jamia Millia Islamia	313	2	5	6	3	12	11	6	6	17	33	46	47	65	54	28.86
6	University of Hyderabad	250	9	10	17	16	8	18	14	23	19	13	14	24	25	40	12.16
7	Pondicherry University	197	3	7	6	10	6	16	5	10	18	18	21	29	20	28	18.75
8	Tezpur University	160	1	2	5	14	18	11	9	11	10	13	12	30	12	12	21.06
9	Babasaheb Bhimrao Ambedkar University	105	0	1	0	0	0	1	1	0	4	9	15	32	26	16	25.99
10	North Eastern Hill University	94	0	0	3	7	2	1	4	2	9	5	11	13	19	18	17.69
11	University of Allahabad	73	3	1	1	0	4	4	5	6	5	5	7	13	10	9	8.82
12	Visva Bharati University	70	0	4	2	2	3	3	7	4	4	7	13	3	10	8	5.95

13	Central University of Rajasthan	64	0	0	1	0	1	0	2	5	4	6	8	10	10	17	29.38
14	Guru Ghasidas Vishwavidyalaya	63	1	1	3	1	1	4	2	2	5	9	3	6	12	13	21.81
15	Assam University	62	1	1	1	2	2	2	2	2	9	4	8	11	7	10	19.38
16	Tripura University	51	0	0	0	1	0	0	1	1	3	9	5	9	10	12	28.21
17	Central University of Jammu	44	0	0	0	0	0	1	0	1	1	2	2	3	10	24	48.77
18	Mizoram University	41	0	1	0	0	0	1	1	2	4	3	9	5	5	10	21.15
19	Central University of Haryana	40	0	0	0	0	0	0	0	0	1	4	0	6	14	15	71.88
20	Sikkim University	35	0	0	0	0	0	0	1	2	4	2	7	9	5	5	25.85
21	Indira Gandhi National Tribal University	35	0	0	0	0	0	0	0	0	0	2	3	12	9	9	45.65
22	Central University of Himachal Pradesh	33	0	0	0	0	0	0	0	0	0	1	11	6	6	9	73.21
23	Rajiv Gandhi University	26	0	1	0	4	4	1	1	5	2	1	2	1	2	2	6.5
24	Central University of Punjab	24	0	0	0	0	0	0	1	2	0	1	2	5	3	10	38.95
25	Dr. Hari Singh Gour University	22	0	0	2	0	1	1	0	4	2	3	2	1	3	3	3.75

26	Central University of South Bihar	19	0	0	0	1	0	0	1	1	1	4	0	2	4	5	17.46
27	Central University of Tamil Nadu	19	0	0	0	0	0	1	0	0	0	1	0	5	6	6	25.1
28	Hemwati Nandan Bahuguna Garhwal University	16	0	0	0	0	1	1	0	0	0	1	1	4	1	7	24.14
29	Central University of Kashmir	11	0	0	0	0	0	0	0	0	1	0	3	2	5	0	49.53
30	Central University of Gujarat	10	0	0	0	0	0	0	0	5	1	1	1	0	1	1	-23.53
31	Central University of Karnataka	7	0	0	0	0	0	0	0	1	0	0	2	0	1	3	20.09
32	Central University of Kerala	6	0	0	0	0	0	0	2	0	0	1	0	1	1	1	-9.43
33	Manipur University	6	0	1	0	0	0	1	0	0	1	0	0	1	0	2	5.95
34	Central University of Jharkhand	4	0	0	0	0	0	0	0	0	0	0	1	2	0	1	0
35	Nagaland University	3	0	0	0	0	0	0	0	0	0	0	1	0	1	1	0
36	English & Foreign Languages University	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	NA

4.2 Analysis of thematic trajectories of research in the 7 computer science categories of Web of Science

Figures 3 and 4 illustrate a consistent rise in the number of publications between 1999 and 2023, with a significant surge observed from 2015 to 2022. This trend highlights the remarkable advancements in computer science research over the past few decades.

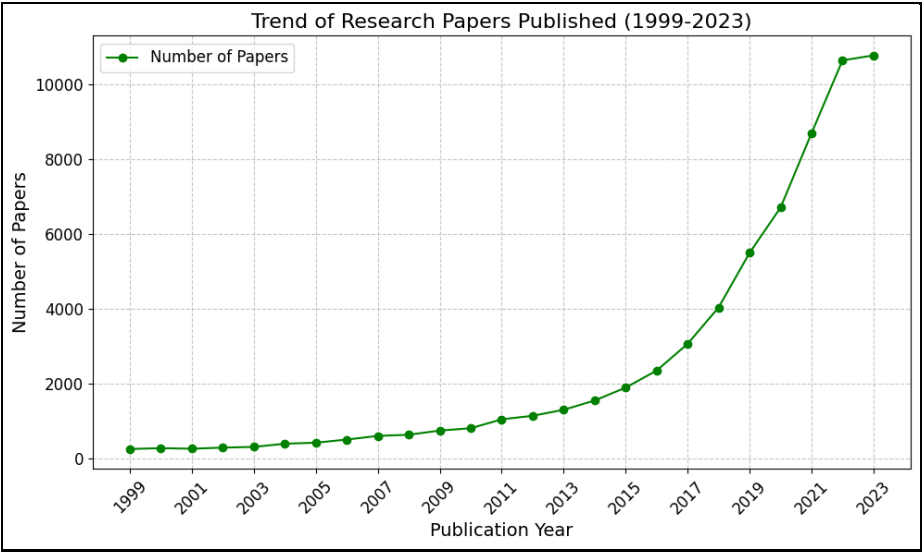
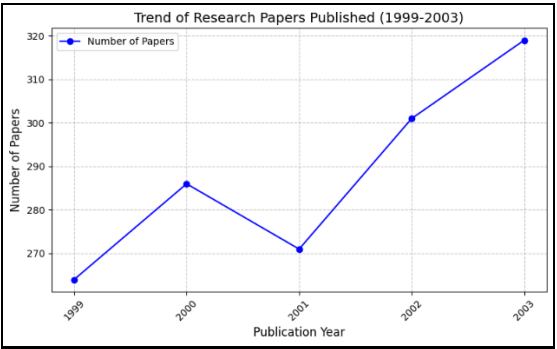
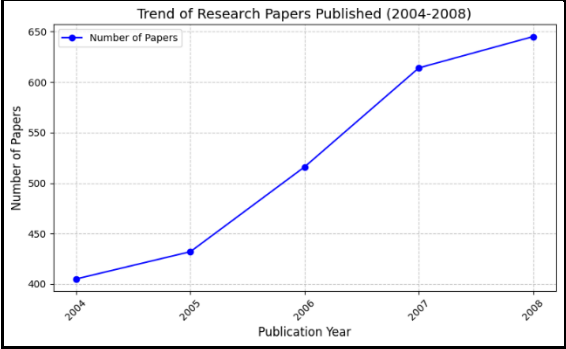


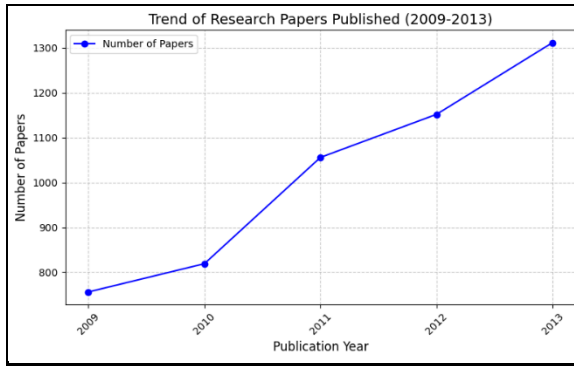
Figure 3: Volume of India’s Computer Science Research Output during 1999-2023



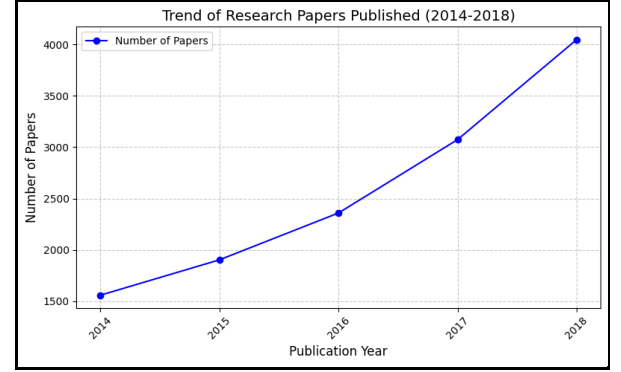
(a)



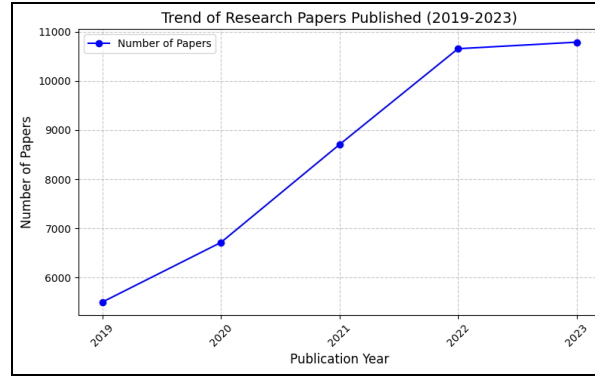
(b)



(c)



(d)



(e)

Figure 4: (a), (b), (c), (d) and (e) depict the Volume of India's Computer Science Research Output during the periods 1999-2003, 2004-2008, 2009-2013, 2014-2018 and 2019-2023 respectively.

From Figure 4, it can be observed that the rate of increase in the number of research papers increased from the period of 1999-2003 to 2019-2023. The rapid increase in the rate of publication could be seen in the recent decade of 2013 to 2023, which exhibited greater emphasis on research-oriented activities in recent times.

4.2.1 Burst detection results of individual periods

In **Table 3**, that depicts the bursty keywords for the overall period of 1999-2023, the words like “deep”, “learn”, “convolut”, “cnn” etc. illustrate the bursty keywords which are the root words of the terminologies like, Deep Learning (DL), Convolutional Neural Network (CNN) and so on. Owing to the maximum number of publications in the recent decade and most prevalent terms in these publications, terms like DL, CNN, COVID-19, Blockchain etc. show burstiness in Author keywords of this period.

According to a burst detection algorithm, the tabular data shows significant trends in keyword usage in computer science research between 1999 and 2023, which is represented graphically in **Figure 5**. Cutting-edge technologies, such as “deep learning”, “convolutional neural networks (CNN)”, “blockchain”, “COVID-19”, “IoT”, and “LSTM”, are heavily emphasized in recent years (2021–2023), reflecting developments in artificial intelligence,

healthcare, and the Internet of Things. Prior eras (1999–2015) concentrated on fundamental subjects such as “genetic algorithms”, “simulation”, “finite element methods”, and “QSAR”, suggesting a transition in subsequent years from foundational research to more application-driven and transformative fields. New words like “adversarial”, “federated”, and “healthcare” draw attention to how AI is increasingly being used in security, team learning, and medical applications.

Table 3: Top 30 words in keywords analysed by burst detection algorithm during the overall period of 1999-2023

Word	Level	Weight	Length	Start	End
Deep	1	717.9374	3	2022	2023
Learn	1	588.0581	3	2022	2023
Convolute	1	240.8413	3	2022	2023
Cnn	1	160.444	3	2022	2023
covid-19	1	141.5352	4	2021	2023
blockchain	1	128.7808	3	2022	2023
genet	1	114.4351	17	1999	2015
attent	1	93.17859	3	2022	2023
lstm	1	83.45094	3	2022	2023
thing	1	82.521	4	2021	2023
diseas	1	77.18186	2	2023	2023
simul	1	66.82146	17	1999	2015
design	1	64.7963	11	2003	2013
dock	1	62.57003	10	2008	2017
term	1	58.15157	2	2023	2023
adversari	1	56.17706	3	2022	2023
program	1	55.93272	16	1999	2014
internet	1	55.69358	4	2021	2023
qsar	1	51.85996	13	2003	2015

short	1	51.27968	3	2022	2023
iot	1	50.72378	3	2022	2023
long	1	50.12818	3	2022	2023
finit	1	49.55255	12	1999	2010
ensembl	1	48.97051	2	2023	2023
pharmacophor	1	46.89389	12	2006	2017
feder	1	43.7901	2	2023	2023
healthcar	1	43.48997	3	2022	2023
evolut	1	41.84629	8	2009	2016
homolog	1	41.09131	13	2003	2015
vehicl	1	40.15798	3	2022	2023

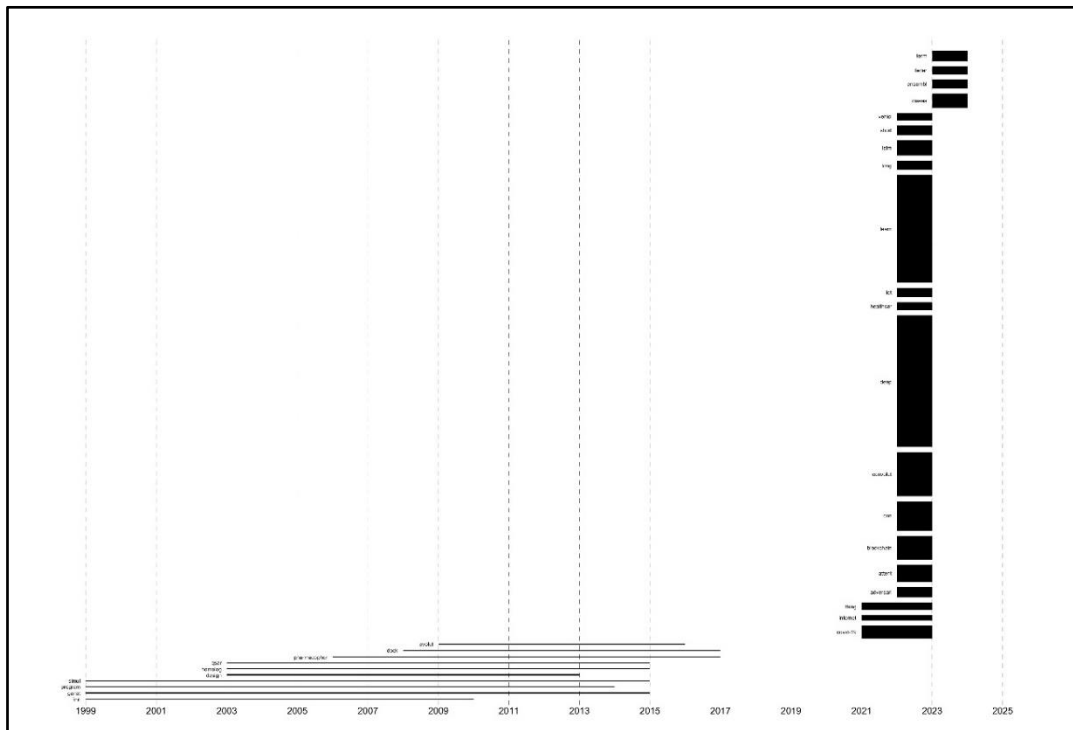


Figure 5: Top 30 keywords extracted by burst detection algorithm for the overall period of 1999-2023

A focus on fundamental ideas in computational research and their applications can be seen in the keywords analyzed between 1999 and 2003, from **Table 4** and **Figure 6**. Important words like “artificial”, “nonlinear”, “virtual”, and “evolution” suggest a focus on early developments in evolutionary computing, nonlinear systems, and artificial intelligence. Terms like “strain”, “systolic”, and “spectrum” imply investigation in the applied sciences, especially in material analysis and signal processing. Words like “fourier”, “observ”, and “textur” draw attention to developments in data representation and image processing. The terms “concurrent”, “continuum”, and “dynam” indicate that system dynamics, concurrency, and parameter optimization were still of interest during this time. All things considered, this dataset captures a time of developing fundamental techniques and investigating cutting-edge computational strategies.

Table 4: Top 30 words in keywords analysed by burst detection algorithm (1999-2003)

Word	Level	Weight	Length	Start	End
line	1	3.211814	1	1999	1999
theori	1	2.617073	2	2002	2003
textur	1	2.591377	1	2003	2003
concurr	1	2.42979	1	2000	2000
artifici	1	2.409559	2	1999	2000
nonlinear	1	2.394878	2	2002	2003
inform	1	2.375796	1	2003	2003
fourier	1	2.348089	1	2003	2003
strain	1	2.312294	1	2000	2000
systol	1	2.311625	2	1999	2000
evolut	1	2.297169	1	2001	2001
observ	1	2.25424	2	2002	2003
virtual	1	2.121787	1	1999	1999
sequenc	1	2.119032	1	2003	2003
spectrum	1	2.002139	1	2000	2000
closur	1	2.002139	1	2000	2000
continuu	1	1.960421	1	2000	2000
compon	1	1.91966	3	2001	2003

stereo	1	1.917707	2	2002	2003
dynam	1	1.907935	2	2002	2003
manag	1	1.879768	1	2000	2000
reliabl	1	1.863597	1	2001	2001
paramet	1	1.857339	2	2002	2003
array	1	1.844194	1	2000	2000
node	1	1.839078	1	2001	2001
iter	1	1.808254	1	2000	2000

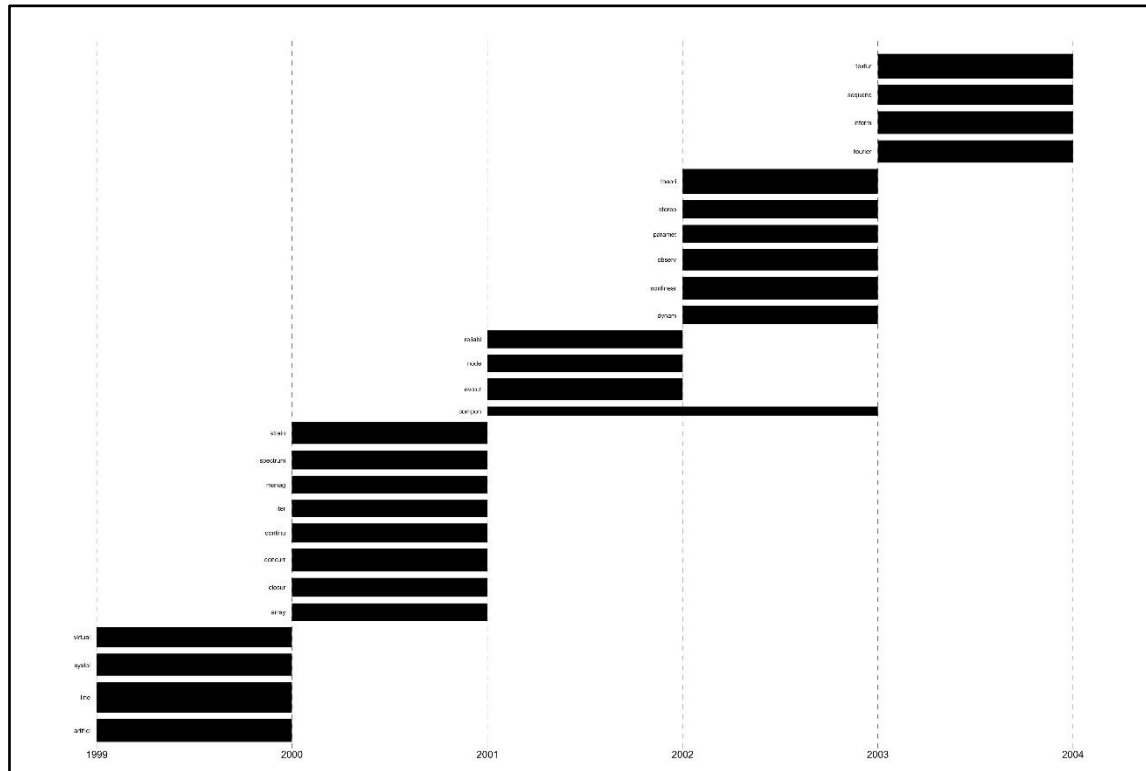


Figure 6: Top 30 keywords extracted by burst detection algorithm (1999-2003)

From **Table 5** and **Figure 7**, the 2004–2008 keyword analysis reveals important developments in computational research and its wide range of applications. Particularly in 2008, terms like “swarm”, “particl”, and “manet” reflect the growing interest in mobile ad hoc networks, particle swarm optimization, and swarm intelligence. Keywords like “wireless” and “power” suggest an emphasis on power efficiency and wireless communication, which were crucial for the advancement of technology at the time. The

terms “pharmacophor” and “medic” indicate advancements in pharmacological research and computational drug discovery. Furthermore, words like “comfa” and “comsia” show how molecular modelling techniques are used. The terms “simul”, “error”, and “predict” highlight developments in predictive modelling, error analysis, and simulation. All things considered, this era shows how specialized fields of study have emerged and how computational methods have been applied to practical issues.

Table 5: Top 30 words in keywords analysed by burst detection algorithm (2004-2008)

Word	Level	Weight	Length	Start	End
swarm	1	6.109878	1	2008	2008
wireless	1	5.404527	1	2008	2008
manet	1	3.859208	1	2008	2008
activ	1	3.825999	1	2004	2004
singl	1	3.669441	1	2005	2005
infer	1	3.600956	1	2008	2008
program	1	3.305421	1	2006	2006
pharmacophor	1	3.265233	1	2008	2008
particl	1	3.109715	1	2008	2008
simul	1	3.093015	1	2005	2005
comfa	1	2.934005	2	2007	2008
length	1	2.902006	1	2008	2008
power	1	2.551717	1	2008	2008
factor	1	2.520553	1	2006	2006
predict	1	2.449587	1	2008	2008
domin	1	2.431909	1	2005	2005
error	1	2.421357	2	2007	2008
circuit	1	2.388593	2	2005	2006
hash	1	2.383974	1	2005	2005
cost	1	2.364186	1	2005	2005
output	1	2.299022	1	2006	2006
medic	1	2.279878	2	2007	2008

exponenti	1	2.272527	1	2004	2004
group	1	2.272527	1	2004	2004
comsia	1	2.256198	2	2007	2008
wave	1	2.221323	1	2005	2005
depend	1	2.215599	1	2006	2006
tool	1	2.203057	1	2008	2008
interpret	1	2.199385	1	2004	2004
case	1	2.15173	1	2004	2004

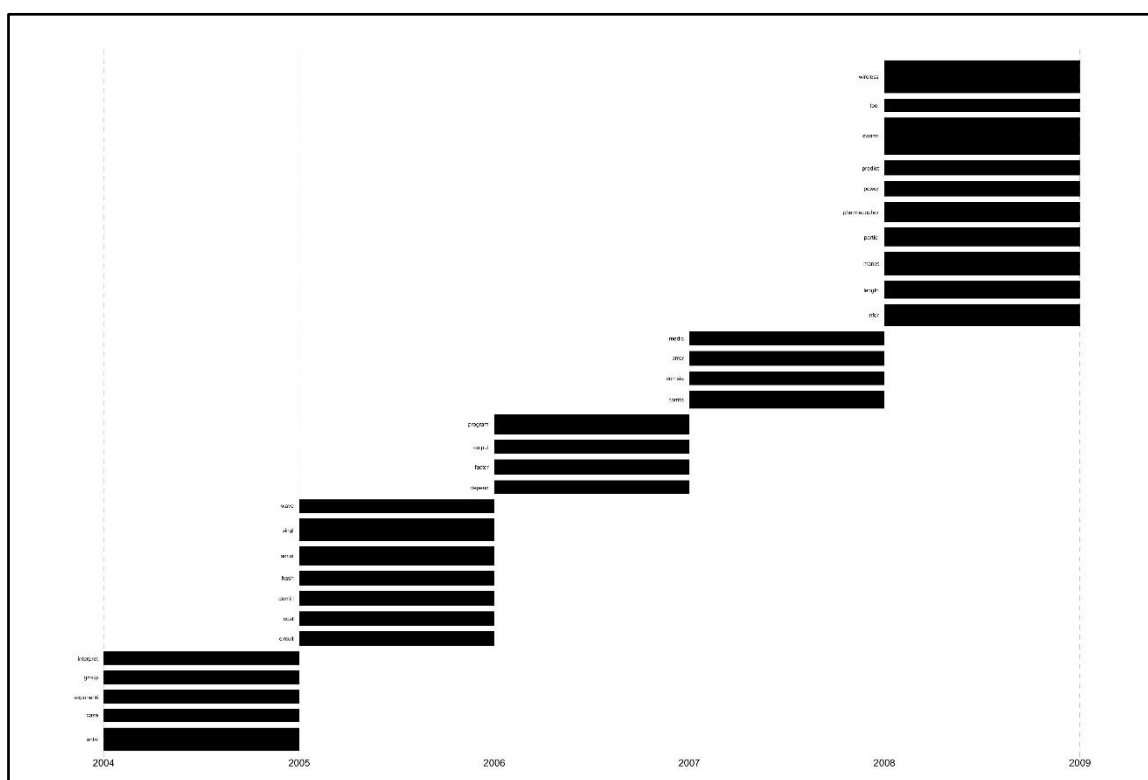


Figure 7: Top 30 keywords extracted by burst detection algorithm (2004-2008)

As can be seen in **Table 6** and **Figure 8**, the keyword analysis for the period 2009–2013 reveals significant trends and emerging areas of research. Keywords like “attribut”, “identif”, and “optimis” illustrate developments in decision-making processes, optimization strategies, and attribute identification. The terms “molecular” and “md” (presumably referring to molecular dynamics) reflect the growing interest in molecular-level simulations and computational biology. Terms like “storag”, “relay”, and “thermal” highlight improvements in infrastructure and system efficiency by referring to advancements in thermal analysis, relay networks, and storage systems. The popularity of the terms “bee”, “hybrid”, and “hoc”

suggests that swarm intelligence, hybrid systems, and ad hoc networks are being studied. Words like “risk”, “recur”, and “triangular” highlight uses in computational geometry, recurrent systems, and risk assessment. All things considered, this era shows how computational research has become more diverse, combining fundamental methods with applications in system design, biology, and emerging technologies.

Table 6: Top 30 words in keywords analysed by burst detection algorithm (2009-2013)

Word	Level	Weight	Length	Start	End
attribut	1	3.908849	1	2011	2011
identif	1	3.780634	1	2011	2011
divis	1	3.703722	1	2009	2009
basi	1	3.618913	2	2009	2010
md	1	3.315337	2	2012	2013
point	1	3.303877	1	2009	2009
associ	1	3.297929	1	2013	2013
storag	1	3.275067	3	2011	2013
bee	1	3.176522	1	2013	2013
techniqu	1	3.079684	1	2013	2013
risk	1	3.055332	1	2013	2013
biolog	1	2.899943	1	2009	2009
optimis	1	2.876093	1	2010	2010
recurr	1	2.746451	1	2013	2013
anchor	1	2.730805	1	2009	2009
h	1	2.712508	1	2010	2010
indic	1	2.661864	2	2012	2013
relay	1	2.625998	1	2013	2013
thermal	1	2.618739	2	2012	2013
hoc	1	2.604589	1	2010	2010
ratio	1	2.600111	1	2009	2009
molecular	1	2.596266	2	2012	2013

hybrid	1	2.580793	1	2010	2010
triangular	1	2.57274	1	2013	2013
overlay	1	2.500031	1	2011	2011
polar	1	2.443583	2	2009	2010
hybrid	1	2.405576	1	2013	2013
summar	1	2.38574	1	2013	2013
independ	1	2.350964	2	2009	2010
hash	1	2.303733	1	2013	2013

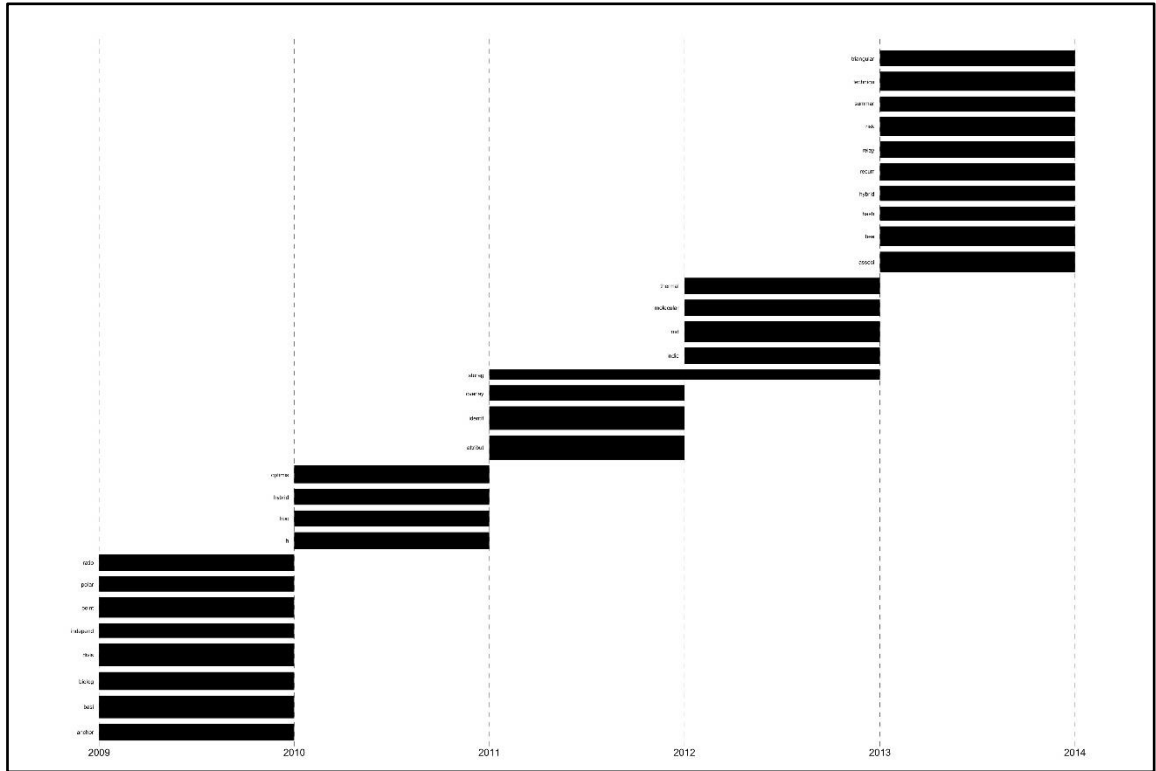


Figure 8: Top 30 keywords extracted by burst detection algorithm (2009-2013)

Important trends in new technologies and research areas are revealed by the keyword analysis for the years 2014–2018, as portrayed in **Table 7** and **Figure 9**. Words like “thing”, “iot”, “internet”, and “fog” emphasize the rapid expansion of fog computing and the Internet of Things (IoT), highlighting developments in decentralized processing and connected devices. The terms “deep”, “cnn”, and “convolut” all denote the growing emphasis on convolutional neural networks and deep learning, which is indicative of their extensive use in a variety of applications. New fields like “big”, “microgrid”, and “health” highlight

developments in sustainable energy, big data analytics, and medical technology. Furthermore, “citi” and “wolf” refer to advancements in urban technology and optimization algorithms. Terms such as “multimedia,” “geometri,” and “scientometr” denote advancements in computational geometry, bibliometric analysis, and multimedia systems. The use of words like “india,” “electroencephalographi,” and “flower” indicates that the scope of the research has expanded to include biological modeling, neurocomputing, and regional studies. All things considered, this era shows a move toward interdisciplinary research, fusing state-of-the-art technologies with useful applications across multiple fields.

Table 7: Top 30 words in keywords analysed by burst detection algorithm (2014-2018)

Word	Level	Weight	Length	Start	End
thing	1	19.35804	2	2018	2018
Deep	1	17.06411	2	2018	2018
internet	1	16.50427	2	2018	2018
Iot	1	16.16209	2	2018	2018
Big	1	8.883907	2	2018	2018
Flow	1	8.068931	1	2014	2014
research	1	6.401625	1	2015	2015
Health	1	5.556388	2	2018	2018
Fog	1	5.047193	2	2018	2018
multimedia	1	4.693588	1	2017	2017
geometri	1	4.681183	1	2015	2015
homolog	1	4.44796	2	2014	2015
microgrid	1	4.415181	2	2018	2018
cnn	1	4.413931	2	2018	2018
life	1	4.202428	1	2015	2015
kinas	1	4.164116	1	2014	2014
citi	1	4.161601	2	2018	2018
anfi	1	4.079433	2	2014	2015
inventori	1	3.984435	1	2015	2015
scientometr	1	3.870881	1	2015	2015

federated learning and decentralized systems for safe data processing grows, the terms "feder" and "blockchain" have been added. With keywords like "covid-19", "sar", "cov-2", and "pandem", which reflect the worldwide scientific response to the COVID-19 pandemic, healthcare and pandemic-related research are well represented. The words "diseas" and "explain" imply a focus on disease modeling and healthcare applications of explainable AI. The introduction of new technologies like "metavers" and "social" also indicates the investigation of virtual worlds and their effects on society. While terms like "badger" and "hate" emphasize the importance of preventing harmful online content, terms like "fake" and "news" draw attention to worries about disinformation and how to spot it. The terms "antenna", "parallel", and "region", among others, imply advancements in geographic studies, parallel computing, and telecommunications. This time frame illustrates the dynamic interaction between tackling pressing global issues like the pandemic and working to develop both established and cutting-edge technologies.

Table 8: Top 30 words in keywords analysed by burst detection algorithm (2019-2023)

Word	Level	Weight	Length	Start	End
attent	1	43.0033	2	2023	2023
feder	1	29.07885	2	2023	2023
explain	1	21.0463	2	2023	2023
adversari	1	18.90536	2	2023	2023
cov-2	1	14.83646	1	2021	2021
ai	1	14.41242	2	2023	2023
diseas	1	13.55278	2	2023	2023
manet	1	12.84035	1	2019	2019
sar	1	12.79153	1	2021	2021
deep	1	12.50044	2	2023	2023
covid-19	1	11.08521	2	2021	2022
pandem	1	10.98555	1	2021	2021
antenna	1	10.68142	1	2019	2019
licens	1	10.19541	1	2021	2021
short	1	10.16794	1	2024	2023
silicon	1	9.662972	1	2021	2021
metavers	1	9.508516	2	2023	2023

match	1	9.448677	2	2019	2020
parallel	1	9.387348	1	2019	2019
u	1	9.364948	2	2023	2023
colour	1	9.222297	2	2019	2020
region	1	9.166227	1	2020	2020
fake	1	8.919776	1	2024	2023
blockchain	1	8.896413	2	2023	2023
badger	1	8.880385	2	2023	2023
hate	1	8.749187	2	2023	2023
bert	1	8.668697	2	2023	2023
social	1	8.658288	1	2024	2023
news	1	8.618383	2	2023	2023
avispa	1	8.537942	2	2019	2020

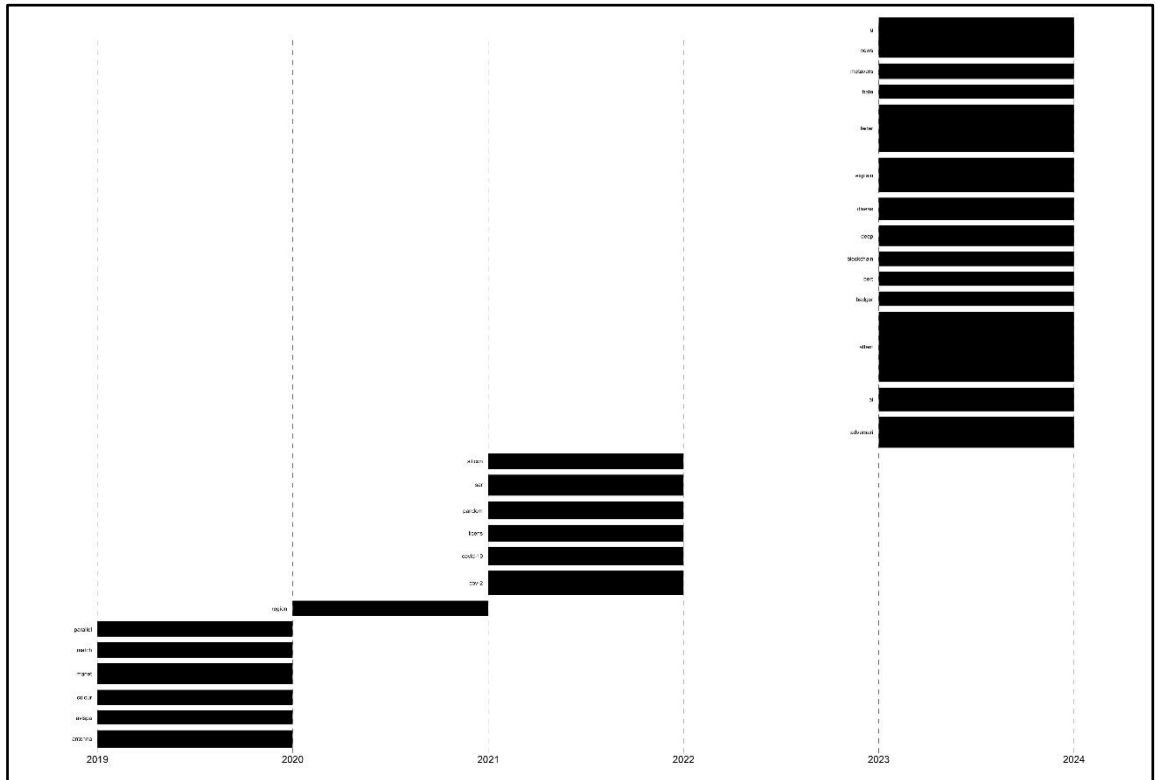


Figure 10: Top 30 keywords extracted by burst detection algorithm (2019-2023)

4.2.2 Clustering results

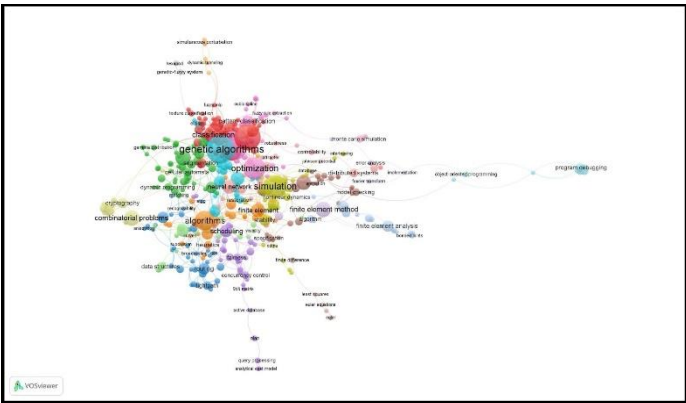
Figure 11 and **Table 9**, offer a study of keyword trends over time, revealing a clear shift in research focus from fundamental computational techniques to advanced artificial intelligence and modern technology applications. From 1999 to 20023 the primary focus was on “fuzzy systems”, “optimization”, “analysis”, and “network algorithms”, with research tackling fundamental issues in modelling, combinatorial problems, and finite element approaches. Cluster analysis emphasizes the importance of “genetic algorithms” and “classification” as key methods for resolving complicated engineering and mathematics problems. This period laid the groundwork for computational intelligence by developing the key frameworks that would support future advances. Since 2004, research has shifted toward applied fields, with a focus on “data mining”, “performance evaluation”, and “neural networks”. By 2014-2018, terms like “cloud computing”, “tensor networks”, and “global optimization” have evolved, indicating the incorporation of computational approaches into real-world applications such as molecular docking, security, and clustering. The era from 2019 through 2023 saw a paradigm shift toward cutting-edge technology such as “deep learning”, “convolutional neural networks”, “blockchain”, and the “Internet of Things”. These breakthroughs highlight the importance of AI-driven solutions to communication system concerns such as energy efficiency and system stability. Overall, the trend reflects the constant evolution of research goals, from theoretical underpinnings to meeting current technology demands.

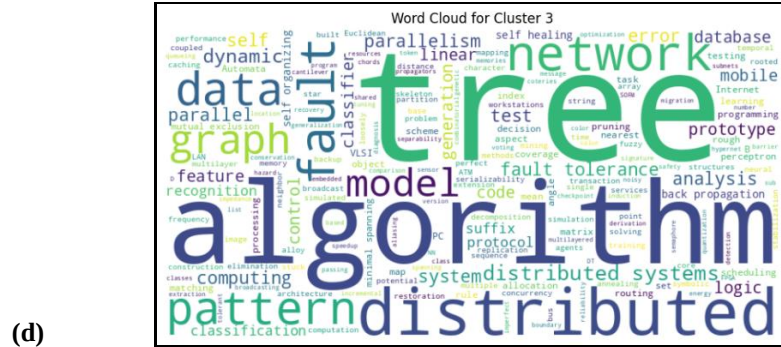
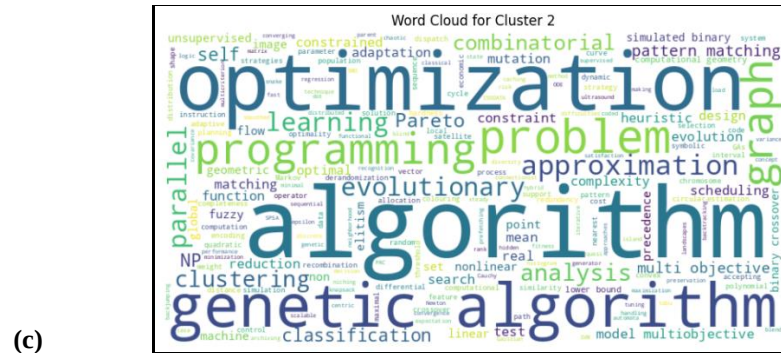
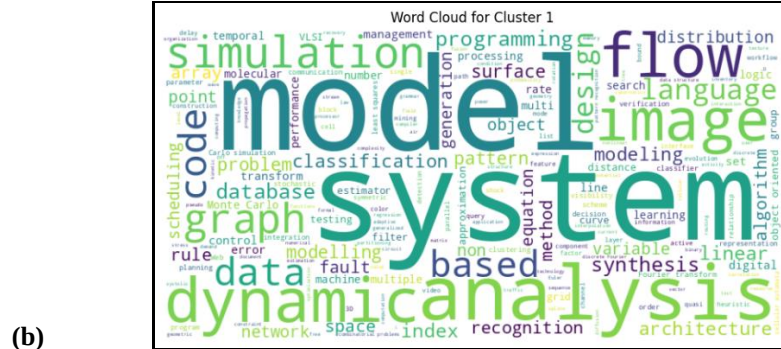
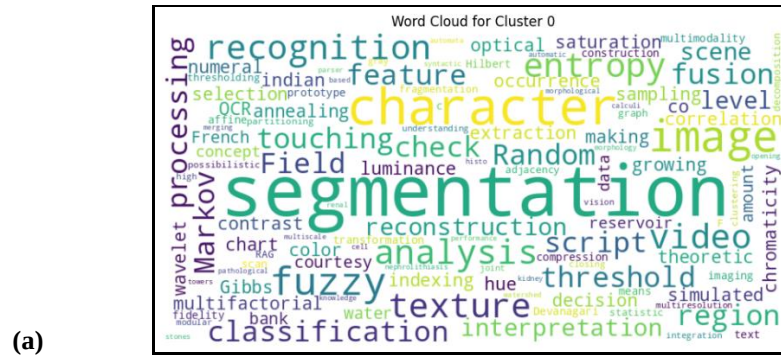
Clusters’ representation for time periods 1999-2003 and 2019-2023

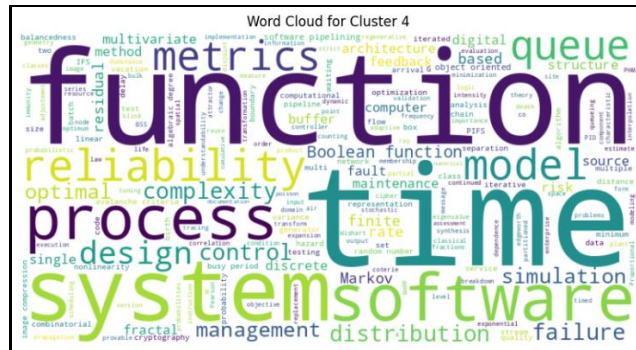
Period

Cluster

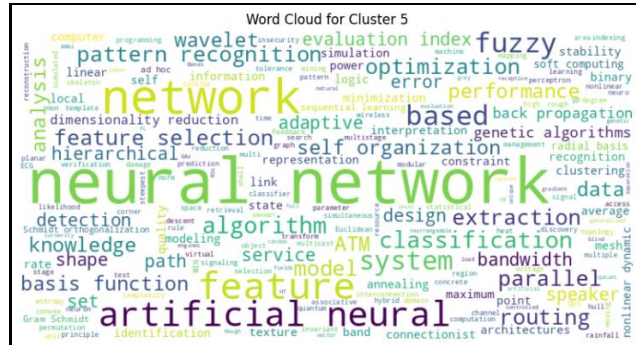
1999-2003



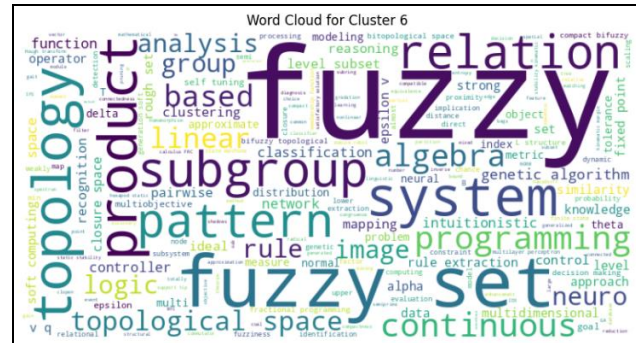




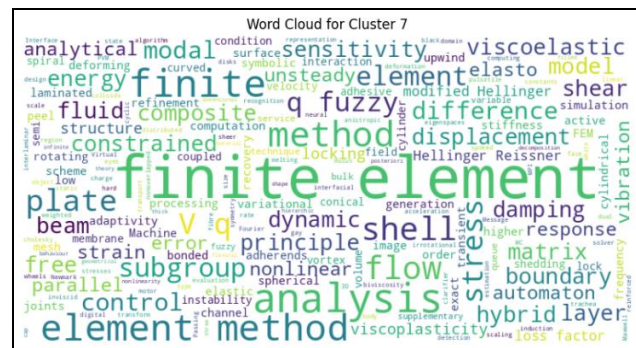
(e)



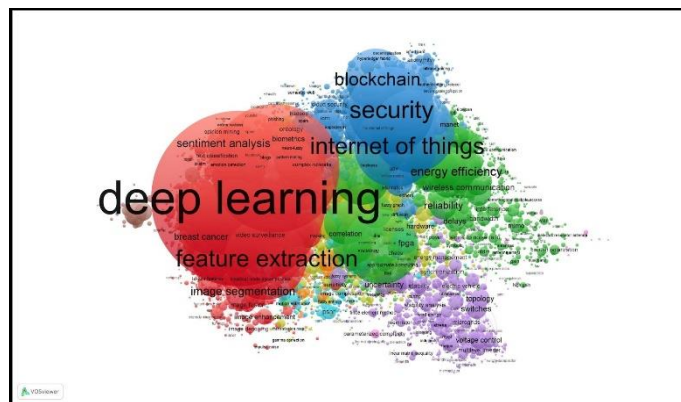
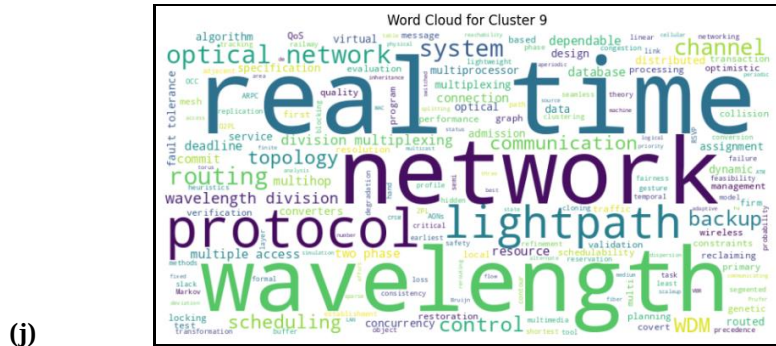
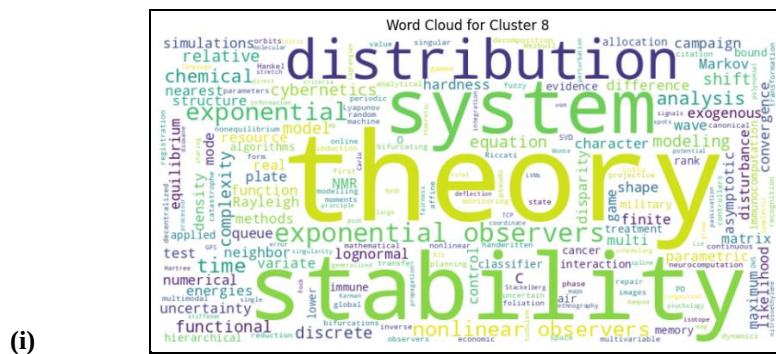
(f)



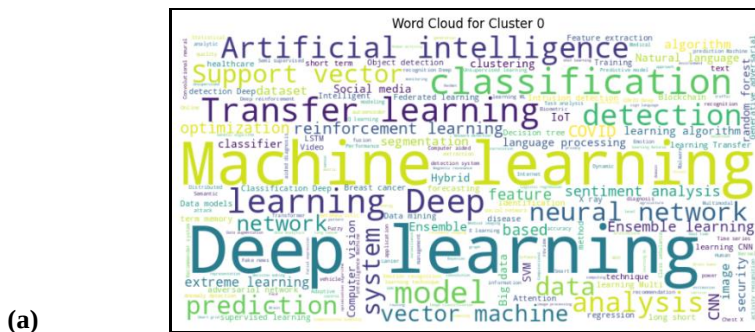
(g)

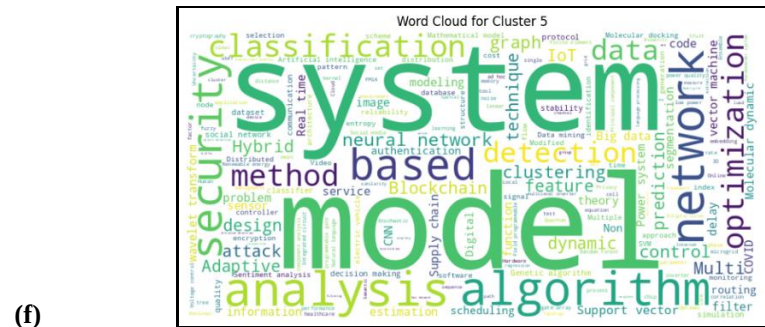
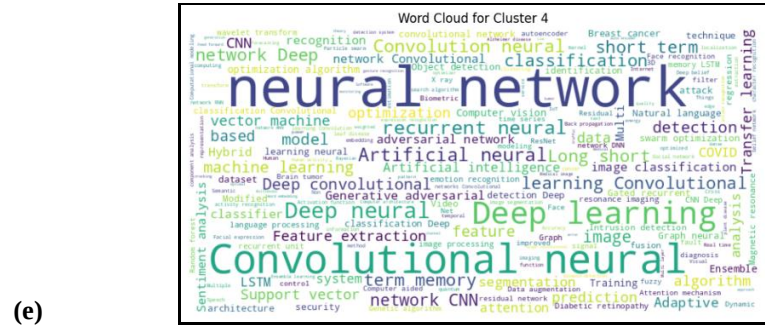
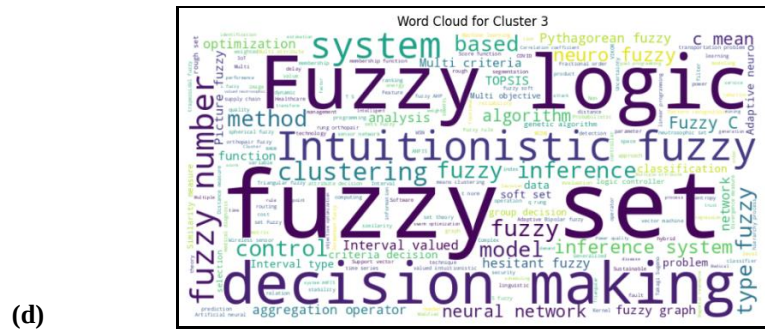
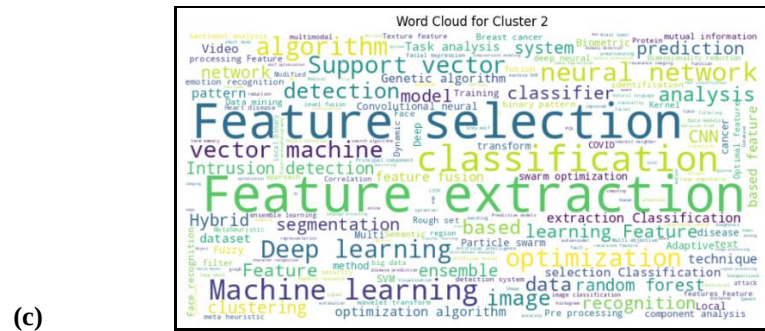
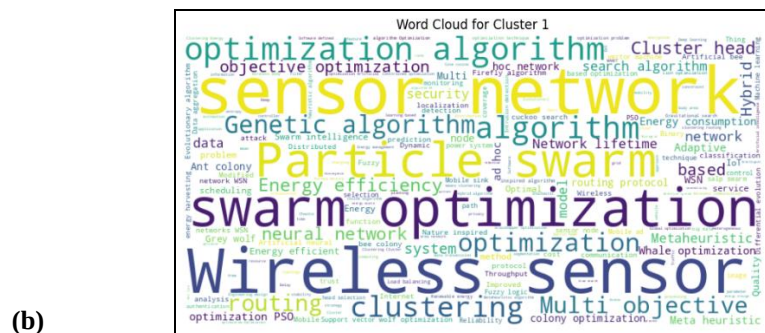


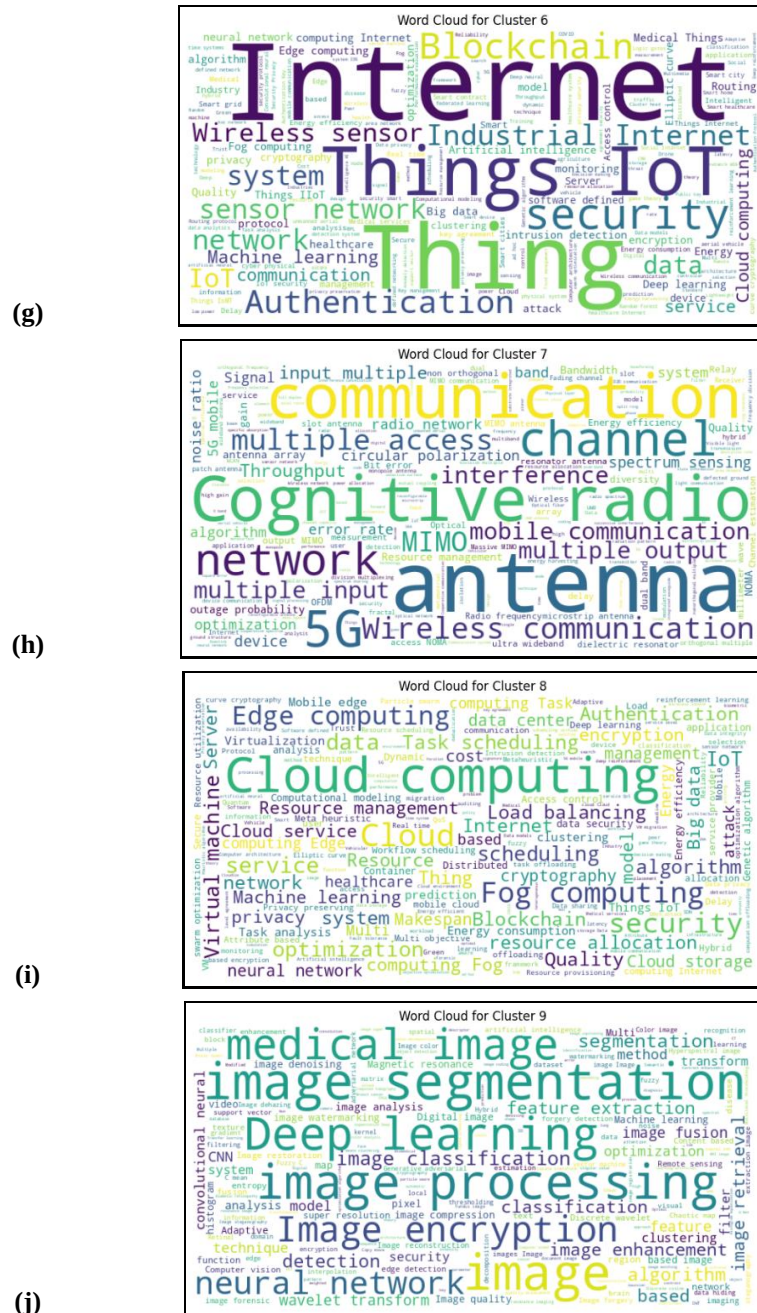
(h)



2019-2023







	network, algorithm, optimisation	optimisation; algorithms; scheduling; combinatorial problems; program debugging; finite element method; data structure	(b)	model, system, analysis, flow, dynamic, simulation
			(c)	algorithm, optimisation, genetic algorithm, programming problem
			(d)	tree, algorithm, distributed, pattern, fault, data, graph, network
			(e)	function, time, system, process, software, reliability, metrics, queue, model
			(f)	neural network, artificial neural, network
			(g)	fuzzy, fuzzy set, topology, relation, product, subgroup
			(h)	finite element, analysis, element method, viscoelastic, shell
			(i)	theory, stability, system, distribution
			(j)	real, time, network, wavelength, protocol, lightpath
2004- 2008	system, fuzzy, model, neural network, analysis, algorithm, data, design	genetic algorithm, ; pattern recognition; performance evaluations; qsar, qspr; wavelet transform; nonlinearity; monte carlo simulation; asymptotic distribution, gamma distribution	(a)	analysis, system, function, design, control, modeling, based, graph
			(b)	algorithm, genetic algorithm, optimisation, multi objective, particle swarm
			(c)	system, neural network, fault, control, artificial neural, fault tolerance

			(d)	network, wireless, ad hoc, routing, sensor network
			(e)	support vector machine, pattern, feature, recognition
			(f)	model, method, finite element, analysis
			(g)	fuzzy, fuzzy logic, system, model, classification
			(h)	data, algorithm, data mining, clustering, analysis, design, program
			(i)	image, wavelet, transform
			(j)	network, neural network, artificial neural
2009-2013	neural network, model, system, analysis, network, algorithm, genetic algorithm	genetic algorithm, support vector machine (svm); docking, md simulations;	(a)	data mining, clustering, classification, association rule, information
			(b)	system, model, method, function, design, algorithm
			(c)	neural network, artificial neural, network, basis function, radial bias
			(d)	wireless, sensor, network, ad hoc, mesh network, routing, WSN, protocol
			(e)	analysis, principal, component analysis, regression, segmentation, feature, dynamic, finite element
			(f)	fuzzy, fuzzy set, system fuzzy

				logic, fuzzy number, intuitionistic fuzzy
			(g)	algorithm, performance, genetic algorithm, design, graph, theory, approximation
			(h)	matrix inequality, neural network, linear matrix, delay, varying delay, time varying, inequality lyapunov, system
			(i)	optimisation, particle swarm, multi objective, differential evolution, objective optimisation, ant colony
			(j)	support vector, vector machine, classification
2014-2018	neural network, system, analysis, model, algorithm, sensor network, classification, particle swarm	cloud computing, security, smart card; neural networks, global optimisation; qsar, molecular docking; image segmentation, image retrieval; minimum likelihood estimation, importance sampling, bootstrap confidence	(a)	data, big data, data mining, analysis, clustering
			(b)	particle swarm, swarm optimisation, multi objective, objective optimisation, bee colony
			(c)	image, wavelet transform, discrete wavelet, feature, adaptive, detection, image processing
			(d)	support vector, vector machine, machine learning, classification
			(e)	neural network, artificial neural network

			(f)	feature, feature extraction, classification, image, recognition
			(g)	fuzzy, fuzzy set, intuitionistic fuzzy, fuzzy number, decision making, interval valued
			(h)	system, model, analysis, network method, security
			(i)	cloud computing, virtual machine, authentication
			(j)	wireless network, sensor network, routing, network lifetime, coverage, efficiency
			(k)	algorithm, genetic algorithm, search algorithm, optimization
2019-2023	neural network, deep learning, machine learning, convolutional neural, model, feature extraction, internet, cloud computing	deep learning, feature extraction, sentiment analysis, biometrics, image segmentation, breast cancer; security, internet of things, blockchain, cloud security; energy efficiency, wireless communication, reliability, correlation, fpga; topology, uncertainty; psnr, motion estimation	(a)	deep learning, machine learning, artificial intelligence, transfer learning
			(b)	sensor network, wireless sensor, swarm optimization
			(c)	feature extraction, feature selection
			(d)	fuzzy logic, fuzzy set, decision making, fuzzy number, intuitionistic fuzzy
			(e)	neural network, convolution neural, recurrent neural, artificial neural, deep convolutional
			(f)	system, model, network, security

			(g)	internet, things, IoT, authentication
			(h)	communication, cognitive radio, antenna, 5G, multiple access, interference, MIMO
			(i)	cloud computing, edge computing, cloud
			(j)	image segmentation, image processing, deep learning, image encryption

4.2.3 Results of graphical representation techniques

According to the data as depicted in **Figure 12** and **Table 10**, computational research has clearly advanced over time, beginning with fundamental methods like neural networks, genetic algorithms, and optimization techniques between 1999 and 2003. As time went on, the emphasis shifted from theoretical frameworks to real-world problem-solving, with examples such as “fuzzy logic”, “data mining”, and “combinatorial problems”. Research progressed toward applied fields such as “QSAR”, “pattern recognition”, and “fuzzy logic” between 2004 and 2008, and interdisciplinary cooperation increased to tackle challenging problems. Between 2009 and 2013, advancements in “docking”, “classification”, and “neural networks”, as well as optimization techniques like “particle swarm optimization” and “differential evolution”, emerged, demonstrating an increasing focus on practical applications such as wireless sensor networks. The terms “cloud computing”, “security”, “machine learning”, and “IoT” were heavily emphasized between 2014 and 2018, with particular attention paid to “optimization” and “feature extraction”. During this time, research themes started to incorporate new technologies like “machine learning” and “big data”, while also addressing contemporary issues like “cloud computing” and “IoT”. Beginning in 2019, the emphasis shifted to more sophisticated technologies like “deep learning”, “blockchain”, and “convolutional neural networks”, which reflected the need to address issues with large amounts of data and guarantee safe, decentralized systems. The growing significance of “cloud computing” and “IoT” has also made “security” and “privacy” important research issues. Overall, the data demonstrates a move away from simple computational techniques and toward advanced, application-driven research that

aims to address modern issues like global health crises and the increasing need for scalable, secure technologies.

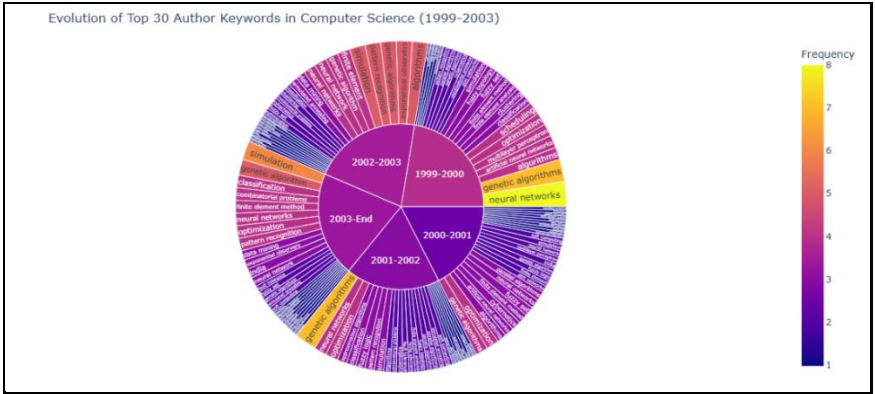
Alluvial Diagram Analysis of Periods of 1999-2003 and 2019-2023

Period

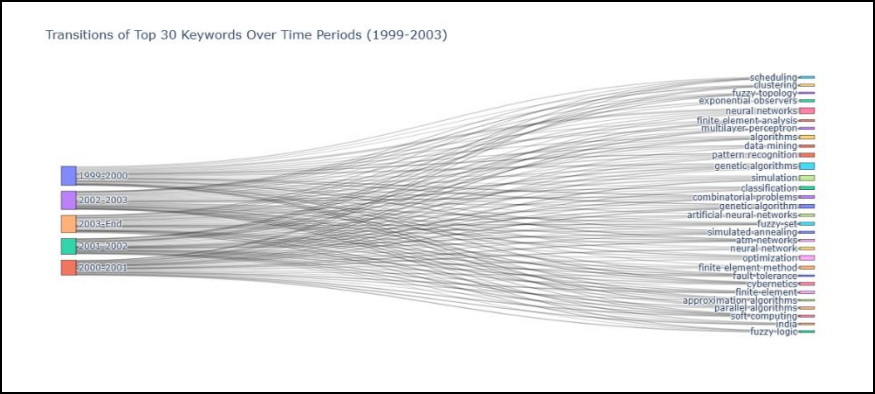
Diagrammatic Representation

1999-2003

(a)

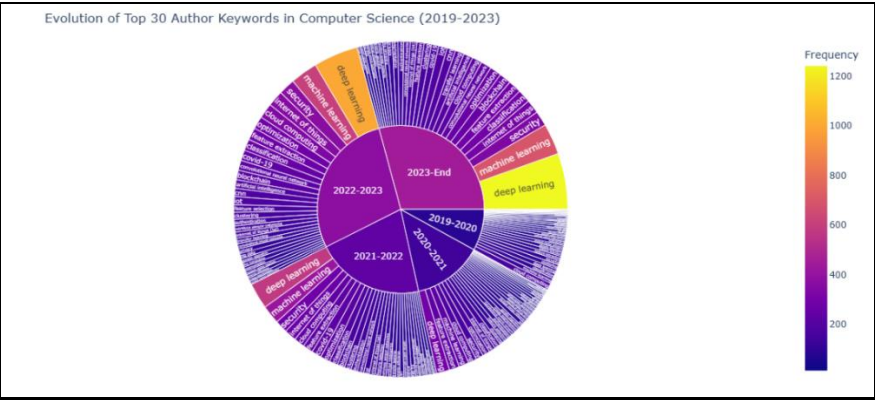


(b)



2019-2023

(a)



(b)

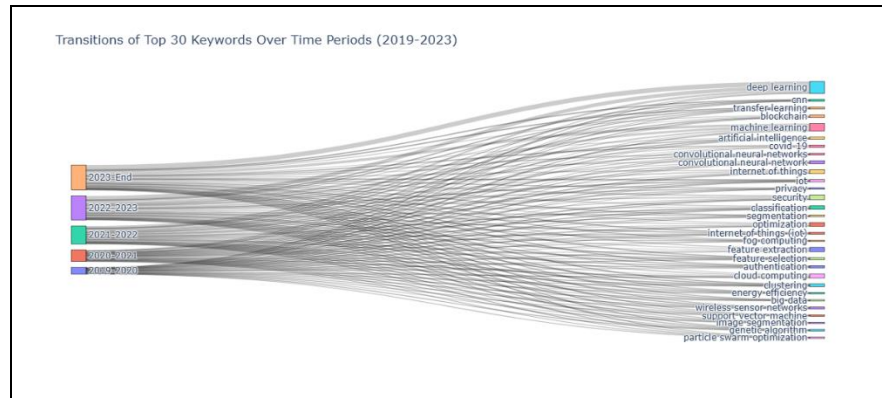


Figure 12: Alluvial / Sankey Diagrams Representing the Evolution Keywords in Different Periods

Table 10: Inference from Alluvial Diagrams

Period	Major Keywords		Inference
	Years	Keywords	
1999-2003	1999-2000	neural networks, genetic algorithms, algorithms, artificial neural networks, multilayer perceptron, optimisation, scheduling, classification, clustering, finite element analysis/method	Between 1999 and 2003, the table shows how computational research progressed from fundamental approaches like neural networks and genetic algorithms to sophisticated applications like fuzzy logic, data mining, and combinatorial problems. This evolution demonstrates a move from theoretical modelling to practical, application-driven approaches to tackling complex problems.
	2000-2001	genetic algorithms, optimisation, algorithms, artificial neural networks, cybernetics, finite element method, fuzzy set, neural networks, parallel algorithms, atm networks	
	2001-2002	genetic algorithms, neural networks, optimisation, approximation algorithms, classification, fuzzy logic, pattern recognition, simulation, algorithms, combinatorial problems	
	2002-2003	algorithms, exponential observers, genetic algorithms, pattern recognition, simulation, finite element, neural network, data mining, simulated annealing, classification	
	2003-End	simulation, genetic algorithm, classification, combinatorial problems, finite element method, neural networks, optimisation, pattern recognition, data mining, exponential observers	

2004-2008	2004-2005	qsar, algorithms, classification, fuzzy logic, genetic algorithm, pattern recognition, performance evaluation, soft computing, design, fuzzy sets	Between 2004 and 2008, computational research shifted significantly toward practical areas, with emphasis on fuzzy logic, data mining, QSAR, and pattern recognition. This period saw advances in feature selection, soft computing, and design optimization, indicating a shift toward more application-driven approaches in computer systems. The use of approaches such as CoMFA, 3D-QSAR, and ad hoc networks demonstrates interdisciplinary collaboration aimed at addressing practical and challenging challenges.
	2005-2006	simulation, genetic algorithms, neural networks, data mining, optimisation, pattern recognition, fuzzy logic, performance, rough sets, ad hoc networks	
	2006-2007	genetic algorithm, qsar, optimisation, design, algorithms, artificial neural networks, clustering, classification, pattern recognition, soft computing	
	2007-2008	genetic algorithm, neural network, algorithms, clustering, CoMFA, data mining, optimisation, 3D-QSAR, performance, performance evaluation	
	2008-End	genetic algorithm, neural network, fuzzy logic, algorithms, QSAR, artificial neural network, CoMFA, data mining, design, reliability	
2009-2013	2009-2010	genetic algorithm, algorithms, clustering, design, fuzzy logic, neural network, artificial neural network, particle swarm optimisation, differential evolution, docking	Between 2009 and 2013, researchers focused on genetic algorithms, particle swarm optimization,
	2010-2011	genetic algorithm, docking, algorithms, artificial neural networks, particle swarm optimisation, neural network, feature selection, qsar, support vector machine, classification	

	2011-2012	genetic algorithm, particle swarm optimisation, docking, differential evolution, classification, optimisation, support vector machine, clustering, fuzzy logic, neural network	and differential evolution, as well as breakthroughs in docking, classification, and neural networks. The focus turned to optimization techniques and their practical applications, such as wireless sensor networks and performance enhancement.
	2012-2013	genetic algorithm, docking, particle swarm optimisation, optimisation, simulation, algorithms, performance, classification, qsar, differential evolution	
	2013-End	genetic algorithm, algorithms, clustering, docking, particle swarm optimisation, fuzzy logic, classification, neural networks, optimisation, performance, wireless sensors	
2014-2018	2014-2015	genetic algorithm, particle swarm optimisation, clustering, differential evolution, cloud computing, feature selection, optimisation, fuzzy logic, wireless sensor networks, molecular dynamics	From 2009 to 2013, research highlighted genetic algorithms, particle swarm optimization, and differential evolution, combined with breakthroughs in docking and classification. The data depicts a shift in research focus from fundamental approaches such as genetic algorithms and clustering (2014-2015) to sophisticated applications in cloud computing, security, and machine learning (2015-2017). By 2018, emergent technologies like IoT and big data have gained
	2015-2016	genetic algorithm, wireless sensor networks, particle swarm optimisation, classification, clustering, feature selection, cloud computing, differential evolution, support vector machine, docking	
	2016-2017	genetic algorithm, cloud computing, optimisation, security, clustering, authentication, machine learning, particle swarm optimisation, optimisation, wireless sensor networks	
	2017-2018	cloud computing, classification, particle swarm optimisation, feature selection, optimisation, security, authentication, clustering, genetic algorithm, feature extraction	
	2018-End	cloud computing, classification, feature extraction, security, genetic algorithm, machine learning, clustering, internet of things, particle swarm optimisation, big data	

			prominence, indicating a shift toward addressing real-world difficulties while maintaining basic themes like optimization, feature extraction, and neural networks. The focus turned to optimization techniques and their practical applications, such as wireless sensor networks and performance augmentation.
2019-2023	2019-2020	cloud computing, machine learning, classification, deep learning, clustering, security, feature extraction, feature selection, genetic algorithm, wireless sensor networks	The data depicts a transition from fundamental approaches like machine learning and clustering (2019-2020) to advanced and emergent subjects like IoT, blockchain, and convolutional neural networks (2022-2023). By 2023, the focus will be on modern difficulties like optimization, cloud computing, and security.
	2020-2021	deep learning, feature extraction, machine learning, security, optimisation, classification, internet of things, cloud computing, feature selection, clustering	
	2021-2022	deep learning, feature extraction, machine learning, cloud computing, security, optimisation, internet of things, classification, clustering, feature selection	
	2022-2023	deep learning, machine learning, security, internet of things, cloud computing, optimisation, feature extraction, classification, COVID-19, convolution neural network, blockchain	
	2023-End	deep learning, machine learning, security, internet of things, classification, feature extraction, blockchain, optimization, convolutional neural network, cloud computing	

Chapter 5

Discussion and Conclusion

In this work, we studied and explored the thematic trends in Computer Science Research for India using standard scientometric techniques and text analytics. We started with data acquisition and obtained broadly two datasets, one for studying the Research Volume of the Indian Central Universities (2010-2023) and another for the Analysis of the Thematic Trajectories in Computer Science (1999-2023). The inferences from the study carried out in this work have been discussed ahead.

5.1 Central universities' research output

The volume of the scholarly data acquired describes that in recent years the number of publications in research articles and review papers has increased, which may be considered a result of more emphasis being laid on the latest computer technologies. The research output of various universities like the Central University of Gujarat, Central University of Haryana, Central University of Himachal Pradesh, Central University of Jharkhand, Central University of Karnataka, Central University of Kashmir, Central University of Kerala, Central University of Punjab, Nagaland University, English and Foreign Languages University and Indira Gandhi National tribal University, those not producing any work in the subject of computer science in the early years of 2010-2015, seem to have started working on the computer science research. The universities like University of Delhi, Banaras Hindu University, Aligarh Muslim University, Jawaharlal Nehru University, University of Allahabad, Jamia Millia Islamia, Pondicherry University, and Tezpur University illustrate a consistently greater number of publications as compared to other universities but they demonstrate comparatively low CAGR.

5.1.1 Compound annual growth rate (CAGR) of research output from Indian central universities

Universities, like UoD (13.47%), BHU (13.18%), AMU (12.88%) etc., with a consistently greater number of publications, exhibit comparatively lower growth in CAGR and the universities with initial negligible publications, like Central Universities of Himachal Pradesh (73.21%), Haryana (71.88%), Karnataka (20.09%) and so on, represent better CAGR in computer science output. The universities like Central University of Himachal Pradesh (73.21%) and Central University of Haryana (71.88%), depict better CAGR rates because their publications in recent years are larger in number as compared to the initial years considered in this study. Universities like Central University of Kerala (-9.43%), and Central University of Gujarat (-23.53%), show negative growth rates, which represent that the number of publications in the later years is less as compared to their publications in the beginning which resulted in the negative CAGR. This represents that CAGR unanimously can't be employed for determining the research output of the institutions. It should be incorporated with some other metric that should cater to both the aspect of consistency and the volume of the research output.

5.2 Analysis of the thematic trajectories in computer science

The dataset for the analysis of thematic trends in the computer science domain was constructed from the Web of Science database, for the period of 1999 to 2023. WoS has seven computer science subject categories, the publications under these categories have been extracted for India and examined further by using various techniques like burst detection, clustering and graphical representational ways like alluvial diagrams.

5.2.1 Integrative view of burst analysis, clustering and graphical representations

Through the burst detection findings, it can be viewed that usage of terminologies like attention mechanism, AI, metaverse, hate, BERT, social, news, deep learning (DL), convolutional neural network (CNN), blockchain, and LSTM are gaining strength in the field of computer science in recent years. This signifies the technological developments in artificial intelligence, the concentration being laid on complex neural architectures, emergence of the real-world applications, devising strategies for addressing societal challenges, constructing scalable and interdisciplinary solutions and focus on human-centric research.

The widespread use of words such as AI, deep learning, BERT (Bidirectional Encoder Representations from Transformers), LSTM (Long Short-Term Memory), and attention mechanism indicates a rapid expansion of machine learning and natural language processing (NLP). These technologies are frequently used for language modelling, text analysis, and sequence prediction, pushing AI's ability to understand human-like situations and meanings. CNN and LSTM are terms that refer to the creation and implementation of advanced neural network models designed for certain applications. CNNs are commonly employed in image processing, computer vision, and feature extraction, whereas LSTMs and attention mechanisms are advancing time-series analysis, sequential data, and NLP jobs. The use of terminology like metaverse and blockchain indicates the intersection between computer science research and emerging technologies. The metaverse represents a growing interest in immersive digital environments, augmented reality (AR), and virtual reality (VR), whereas blockchain emphasizes the necessity of decentralization, security, and trust in fields such as cryptocurrency, supply chain, and data sharing. Keywords such as hatred, social, and news imply an increased focus on combatting disinformation, hate speech, and bias in digital areas. This is consistent with attempts to establish ethical AI systems, increase online safety, and create responsible technology to address critical social concerns. The phrase AI, when paired with blockchain and deep learning, refers to the convergence of several sectors in order to create scalable and efficient solutions. For example, AI-enabled blockchain systems can improve security and efficiency, whereas metaverse AI apps strive to provide more interactive and personalized experiences. The frequent use of terminology like social, news, and hatred shows that computer science is becoming more human-centric, with an emphasis on improving human interactions with technology, reducing societal harm, and improving the general quality of life.

The results from clustering demonstrated that the clusters of keywords evolved from themes like fuzzy theory, optimisation algorithms, neural networks, model and system analysis to cloud computing, deep learning, machine learning, 5G and big data. The analysis of keyword patterns from 1999 to 2023 reveals a gradual transition in computer science research, from basic computational techniques to advanced artificial intelligence and technology-driven applications. The first emphasis (1999-2003) on “fuzzy systems”, “optimization”, and “network algorithms” created the theoretical framework for computational intelligence, with substantial contributions from methods such as “genetic algorithms” and “classification” in tackling engineering and mathematical challenges. As research progressed (2004-2013), the emphasis shifted to practical topics like “data mining”, “performance evaluation”, and “neural networks”, indicating a growing interest in real-world applications. The years 2014-2018 were breakthrough moments, with words like “cloud computing”, “tensor networks”, and “global optimization” emerging to reflect the incorporation of computational methodologies into disciplines such as security, molecular docking, and clustering. From 2019, there was a noticeable paradigm shift, with an increase in interest in cutting-edge technologies like “deep learning”, “blockchain”, “convolutional neural networks”, and the “Internet of Things”. These achievements highlight the growing importance of AI-powered solutions in solving contemporary difficulties such as energy efficiency, system stability, and intelligent communication systems. Overall, this chronology of research evolution depicts a constant path of innovation, from theoretical foundations to the implementation of disruptive technologies that meet current demands.

The study of alluvial diagrams shows the gradual progression of computational research, from foundational techniques to advanced, application-driven solutions. From 1999 to 2003, research centred on foundational approaches such as neural networks, genetic algorithms, and optimization, which laid the platform for computational frameworks. Over time, the emphasis switched to real-world issue solving, as evidenced by a greater emphasis on fuzzy logic, data mining, and combinatorial problems between 2004 and 2008, as well as interdisciplinary collaborations to address complicated challenges. Between 2009 and 2013, advances in docking, classification, and neural networks, as well as optimization approaches like particle swarm optimization and differential evolution, demonstrated a rising emphasis on practical applications, such as wireless sensor networks. From 2014 to 2018, emerging technologies such as machine learning, big data, cloud computing, and IoT were integrated into research to address current difficulties such as optimization, feature extraction, and security. Since 2019, the focus has switched to cutting-edge technologies such as deep learning, blockchain, and convolutional neural networks, motivated by the need to manage massive volumes of data, improve security, and construct decentralized systems. Furthermore, the growing popularity of cloud computing and IoT has highlighted the significance of resolving security and privacy problems. Overall, the findings show a clear path forward in computational research, from fundamental methods to advanced, interdisciplinary approaches to complicated, current concerns.

5.3 Conclusion

Current work offers valuable insights for scholars, academicians and practitioners seeking to understand the historical and current landscape of computer science to guide their future

research pursuits. A scientometric approach, specifically the burst detection method, was employed to identify the most emerging topics in the field of computer science after the application of pre-processing techniques. Additionally, the clustering method followed by graphical representations was applied to the same textual dataset to analyze and understand the research area clusters based on author keywords. The transition from fundamental techniques like neural networks, evolutionary algorithms, and fuzzy systems to sophisticated, application-driven solutions like deep learning, blockchain, and IoT is reflected in the development of computational research between 1999 and 2023. The theoretical frameworks developed by early research (1999–2003) centred on optimization and classification approaches, and between 2004 and 2013, they moved into real-world applications such as data mining, performance evaluation, and neural networks. Innovations from 2014 to 2018 brought new technologies like cloud computing, big data, and machine learning to the real world to solve security, clustering, and optimization problems. The emphasis changed to cutting-edge technologies like blockchain, AI, and CNNs starting in 2019, indicating the need for solutions that are human-centred, scalable, and safe. Keyword trends that emphasize multidisciplinary approaches and the confluence of fields include metaverse, BERT, attention processes, and LSTM. These trends reflect developments in AI-driven technologies, ethical AI, and social problem-solving. This development shows a continuous path of innovation, from theoretical foundations to using cutting-edge technologies to address difficult, contemporary problems.

5.4 Future scope

This work proposes a mechanism for investigating the trending topics in any domain and performs operations to get deep insights into the scientometric techniques and methods. The framework of the methodology employed can be utilised anywhere to determine the trending entities in society. The burst detection method can be utilized to identify the rapid spread of viral misinformation or rumours on social media platforms, a significant challenge in today's digital era, by pinpointing topics that have surged in frequency within a short time frame. This framework could also be incorporated with the latest deep learning and AI technologies to further enhance the output. Further, this work would be elaborated to other scholarly databases like Dimensions, Scopus etc. to get a larger view of the research landscape. Future studies can explore alternative methods for conducting these analyses. Researchers may concentrate on domain-specific studies to offer a detailed view of progress in specific fields. Similarly, regional variations in research trends can be examined to generate country or region-specific insights, assisting scholars and stakeholders in understanding localized developments.

References

1. Baas, J., Schotten, M., Plume, A., Cote, G., & Karimi, R. (2019). Scopus as a curated, high-quality bibliometric data source for academic research in quantitative science studies. *Quantitative Science Studies*, 1(1), 377–386.
2. Basu, A. (2010). Does a country's scientific 'productivity' depend critically on the number of country journals indexed? *Scientometrics*, 82(3), 507-516.
3. Basu, A., Banshal, S. K., Singhal, K., & Singh, V. K. (2016). Designing a composite index for research performance evaluation at the national or regional level: Ranking central universities in India. *Scientometrics*, 107(3), 1171–1193.
4. Bhattacharya, S., Shilpa, & Kaul, A. (2015). Emerging countries assertion in the global publication landscape of science: a case study of India. *Scientometrics*, 103, 387-411.
5. Birkle, C., Pendlebury, A. D., Schnell, J., & Adams, J. (2019). Web of Science as a data source for research on scientific and scholarly activity. *Quantitative Science Studies*, 1(1), 363–376.
6. Cobo, M. J., López-Herrera, A. G., Herrera-Viedma, E., & Herrera, F. (2011). Science mapping software tools: Review, analysis, and cooperative study among tools. *Journal of the American Society for Information Science and Technology*, 62(7), 1382-1402.
7. Collazo-Reyes, F. (2014). Growth of the number of indexed journals of Latin America and the Caribbean: the effect on the impact of each country. *Scientometrics*, 98(1), 197-209.
8. Erfanmanesh, M., Tahira, M., & Abrizah, A. (2017). The publication success of 102 nations in Scopus and the performance of their Scopus-indexed journals. *Publishing Research Quarterly*, 33(4), 421-432.
9. Gusenbauer, M. (2022). Search where you will find most: Comparing the disciplinary coverage of 56 bibliographic databases. *Scientometrics*, 127(5), 2683-2745.
10. Harzing, A. W. (2019). Two new kids on the block: How do Crossref and Dimensions compare with Google Scholar, Microsoft Academic, Scopus and the Web of Science? *Scientometrics*, 120(1), 341–349.
11. Heck, T., Keller, C., & Rittberger, M. (2024). Coverage and similarity of bibliographic databases to find most relevant literature for systematic reviews in education. *International Journal on Digital Libraries*, 25(2), 365-376.
12. Herzog, C., Hook, D., & Konkiel, S. (2020). Dimensions: bringing down barriers between scientometricians and data. *Quantitative Science Studies*, 1(1), 387–395.
13. Hoonlor, A., Szymanski, B. K., & Zaki, M. J. (2013). Trends in computer science research. *Communications of the ACM*, 56(10), 74-83.
14. Jurayeva, Z., & No, S. (2024). THE ROLE OF INFORMATION TECHNOLOGY IN MEDICINE AND BIOMEDICAL ENGINEERING IN TRAINING FUTURE SPECIALISTS DURING THE PERIOD OF DIGITAL TRANSFORMATION IN EDUCATION.
15. Kanaujia, A., Singh, P., Nandy, A., & Singh, V. K. (2022). Research contribution of major centrally funded institution systems of India. *arXiv preprint arXiv:2208.01588*.
16. Kleinberg, J. (2002, July). Bursty and hierarchical structure in streams. In *Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining* (pp. 91-101).
17. Kostelić, K., & Etinger, D. (2024). Securing the Metaverse: A Bibliometric Analysis of Cybersecurity Challenges and Research Trajectories. *IEEE Engineering Management Review*.
18. Leta, J. (2011). Growth of Brazilian science: A real internationalization or a matter of databases' coverage? In *Proceedings of International Conference of the International Society for Scientometrics and Informetrics*, South Africa, (392–408) July 2–7, 2011.
19. Leydesdorff, L., & Milojević, S. (2012). *Scientometrics*. *arXiv preprint arXiv:1208.4566*.
20. Ley, M. (2009). DBLP: some lessons learned. *Proceedings of the VLDB Endowment*, 2(2), 1493-1500.

21. Li, Y., Liu, S., Zeng, A., Wu, J., Zhang, J., Zhang, W., & Li, S. (2024). Interdisciplinary Dynamics in COVID-19 Research: Examining the Role of Computer Science and Collaboration Patterns. *Systems*, 12(4), 113.
22. Lopez-Lopez, D., Plaza-Navas, M. A., Torres-Pruñonosa, J., & Martinez, L. F. (2024). Navigating the landscape of e-commerce: thematic clusters, intellectual turning points, and burst patterns in online reputation management. *Electronic Commerce Research*, 1-38.
23. Mahmoud, M. (2024). Editorial for the Special Issue “Data Science and Big Data in Biology, Physical Science and Engineering”. *Technologies*, 12(1), 8.
24. Marisha, Banshal, S. K., & Singh, V. K. (2017). Research performance of central universities in India. *Current Science*, 112(11), 2198–2207.
25. Martín-Martín, A., Orduna-Malea, E., Thelwall, M., & López-Cózar, E. D. (2018b). Google Scholar, Web of Science, and Scopus: A systematic comparison of citations in 252 subject categories. *Journal of Informetrics*, 12(4): 1160-1177.
26. Martín-Martín, A., Orduna-Malea, E., & López-Cózar, E. D. (2018a). Coverage of highly-cited documents in Google Scholar, Web of Science, and Scopus: a multidisciplinary comparison. *Scientometrics*, 116(3), 2175–2188.
27. Martín-Martín, A., Thelwall, M., Orduna-Malea, E., & Delgado López-Cózar, E. (2021). Google Scholar, Microsoft Academic, Scopus, Dimensions, Web of Science, and OpenCitations’ COCI: a multidisciplinary comparison of coverage via citations. *Scientometrics*, 126(1), 871-906.
28. Michels, C., & Schmoch, U. (2012). The growth of science and database coverage. *Scientometrics*, 93(3), 831-846.
29. Mingers, J., & Lipitakis, E. (2010). Counting the citations: A comparison of Web of Science and Google Scholar in the field of business and management. *Scientometrics*, 85(2): 613-625.
30. Moed, H. F., Markusova, V., & Akoev, M. (2018). Trends in Russian research output indexed in Scopus and Web of Science. *Scientometrics*, 116(2): 1153-1180.
31. Mongeon, P., & Paul-Hus, A. (2016). The journal coverage of Web of Science and Scopus: a comparative analysis. *Scientometrics*, 106, 213-228.
32. Najari, A., & Yousefvand, M. (2013). Scientometrics study of impact of journal indexing on the growth of scientific productions of Iran. *Iranian Journal of public health*, 42(10): 1134.
33. Ortega, J. L. (2014). Academic search engines: A quantitative outlook. Elsevier.
34. Prathap, G. (2014). The performance of research-intensive higher educational institutions in India. *Current Science*, 107(3), 389–396.
35. Rangayyan, R. M., & Krishnan, S. (2024). Biomedical signal analysis. John Wiley & Sons.
36. Rovira, C., Codina, L., Guerrero-Solé, F., & Lopezosa, C. (2019). Ranking by relevance and citation counts, a comparative study: Google Scholar, Microsoft Academic, WoS and Scopus. *Future Internet*, 11(9), 202.
37. Shortliffe, E. H., Shortliffe, E. H., Cimino, J. J., & Cimino, J. J. (2014). Biomedical informatics: computer applications in health care and biomedicine. Springer.
38. Shortliffe, E. H., & Blois, M. S. (2006). The computer meets medicine and biology: emergence of a discipline. *Biomedical informatics: Computer applications in health care and biomedicine*, 3-45.
39. Singh, A., Kanaujia, A., Singh, V. K., & Vinuesa, R. (2023). Artificial intelligence for Sustainable Development Goals: Bibliometric patterns and concept evolution trajectories. *Sustainable Development*, 32(1), 724-754.
40. Singh, T., & Kumari, M. (2021). Burst: real-time events burst detection in social text stream. *The Journal of Supercomputing*, 77(10), 11228-11256.
41. Singh, V. K., Nandy, A., Singh, P., Karmakar, M., Singh, A., Lathabai, H. H., Srichandan, S. S., & Kanaujia, A. (2022a). Indian Science Reports: a web-based scientometric portal for mapping Indian research competencies at overall and institutional levels. *Scientometrics*, 127(7), 4227-4236.
42. Singh, V. K., Singh, P., Karmakar, M., Leta, J., & Mayr, P. (2021). The journal coverage of Web of Science, Scopus and Dimensions: A comparative analysis. *Scientometrics*, 126, 5113-5142.

43. Singh, V. K., Srichandan, S. S., & Lathabai, H. H. (2022b). ResearchGate and Google Scholar: how much do they differ in publications, citations and different metrics and why?. *Scientometrics*, 127(3), 1515-1542.
44. Sohrabi, B., Vanani, I. R., Jalali, S. M. J., & Abedin, E. (2019). Evaluation of research trends in knowledge management: a hybrid analysis through burst detection and text clustering. *Journal of Information & Knowledge Management*, 18(04), 1950043.
45. Tattershall, E., Nenadic, G., & Stevens, R. D. (2020). Detecting bursty terms in computer science research. *Scientometrics*, 122(1), 681-699.
46. Thelwall, M. (2018). Dimensions: a competitor to Scopus and the Web of Science? *Journal of Informetrics*, 12(2), 430–435.
47. Van Noorden, R. (2014). Online collaboration: Scientists and the social network. *Nature news*, 512(7513), 126.
48. Visser, M. S., van Eck, N. J. & Waltman, L. (2019). Large-scale comparison of bibliographic data sources: Web of Science, Scopus, Dimensions and Crossref. In *proceedings of 17th International Conference on Scientometrics and Informetrics*, pp. 2358–2369.
49. Visser, M., Van Eck, N. J., & Waltman, L. (2021). Large-scale comparison of bibliographic data sources: Scopus, Web of Science, Dimensions, Crossref, and Microsoft Academic. *Quantitative science studies*, 2(1), 20-41.

Appendix

Source Code for:

A. Analysis of Research Output of Central Universities

```
import pandas as pd
import numpy as np
data = pd.read_csv("savedrecs.csv")
data.head()
data.dropna(subset=['Author Keywords'], inplace=True)
data["Affiliations"].fillna("Not available", inplace = True)
data
AMU = df[df['Affiliations'].str.contains('Aligarh Muslim University')]
Assam_University = df[df['Affiliations'].str.contains('Assam University')]
BRAmbedkar_University = df[df['Affiliations'].str.contains('Babasaheb Bhimrao Ambedkar University')]
BHU = df[df['Affiliations'].str.contains('Banaras Hindu University')]
CU_Gujarat = df[df['Affiliations'].str.contains('Central University of Gujarat')]
CU_Haryana = df[df['Affiliations'].str.contains('Central University of Haryana')]
CU_Himachal = df[df['Affiliations'].str.contains('Central University of Himachal Pradesh')]
CU_Jammu = df[df['Affiliations'].str.contains('Central University of Jammu')]
CU_Jharkhand = df[df['Affiliations'].str.contains('Central University of Jharkhand')]
CU_Karnataka = df[df['Affiliations'].str.contains('Central University of Karnataka')]
CU_Kashmir = df[df['Affiliations'].str.contains('Central University of Kashmir')]
CU_Kerala = df[df['Affiliations'].str.contains('Central University of Kerala')]
CU_Punjab = df[df['Affiliations'].str.contains('Central University of Punjab')]
CU_Rajasthan = df[df['Affiliations'].str.contains('Central University of Rajasthan')]
CU_S Bihar = df[df['Affiliations'].str.contains('Central University of South Bihar')]
CU_TamilNadu = df[df['Affiliations'].str.contains('Central University of Tamil Nadu')]
HariSinghGour_University = df[df['Affiliations'].str.contains('Dr. Hari Singh Gour University')]
GuruGhasidas_University = df[df['Affiliations'].str.contains('Guru Ghasidas Vishwavidyalaya')]
Hemawati_University = df[df['Affiliations'].str.contains('Hemwati Nandan Bahuguna Garhwal University')]
JMI = df[df['Affiliations'].str.contains('Jamia Millia Islamia')]
JNU = df[df['Affiliations'].str.contains('Jawaharlal Nehru University')]
Manipur_University = df[df['Affiliations'].str.contains('Manipur University')]
Mizoram_University = df[df['Affiliations'].str.contains('Mizoram University')]
Nagaland_University = df[df['Affiliations'].str.contains('Nagaland University')]
NorthEasternHillUniversity = df[df['Affiliations'].str.contains('North Eastern Hill University')]
Pondicherry_University = df[df['Affiliations'].str.contains('Pondicherry University')]
RajivGandhi_University = df[df['Affiliations'].str.contains('Rajiv Gandhi University')]
Sikkim_University = df[df['Affiliations'].str.contains('Sikkim University')]
Tezpur_University = df[df['Affiliations'].str.contains('Tezpur University')]
EngNForLan_University = df[df['Affiliations'].str.contains('English & Foreign Languages University')]
IndiraGandhiTribal_University = df[df['Affiliations'].str.contains('Indira Gandhi National Tribal University')]
Tripura_University = df[df['Affiliations'].str.contains('Tripura University')]
AU = df[df['Affiliations'].str.contains('University of Allahabad')]
DU = df[df['Affiliations'].str.contains('University of Delhi')]
HU = df[df['Affiliations'].str.contains('University of Hyderabad')]
ViswaBharti_University = df[df['Affiliations'].str.contains('Visva Bharati University')]
#Universities with no Research Paper in WoS
CentralSanskrit = df[df['Affiliations'].str.contains('Central Sanskrit University')]
CentralTribalAndhraPradesh = df[df['Affiliations'].str.contains('Central Tribal University of Andhra Pradesh')]
```

```

CentralUnivAndhraPradesh = df[df['Affiliations'].str.contains('Central University of Andhra Pradesh')]
CentralUnivOdisha = df[df['Affiliations'].str.contains('Central University of Odisha')]
MKGHindiVishwavidyalaya = df[df['Affiliations'].str.contains('Mahatma Gandhi Antarrashtriya Hindi Vishwavidyalaya')]
MGCentralUniv = df[df['Affiliations'].str.contains('Mahatma Gandhi Central University')]
UrduUniv = df[df['Affiliations'].str.contains('Maulana Azad National Urdu University')]
NationalSanskritUniv = df[df['Affiliations'].str.contains('National Sanskrit University')]
LalBahadurSanskrit = df[df['Affiliations'].str.contains('Shri Lal Bahadur Shastri National Sanskrit University')]
SindhuUniv = df[df['Affiliations'].str.contains('Sindhu Central University')]
AMU.groupby('Publication Year').count()
AMU.groupby('Publication Year').count()
Assam_University.groupby('Publication Year').count()
BRAmbedkar_University.groupby('Publication Year').count()
BHU.groupby('Publication Year').count()
CU_Gujarat.groupby('Publication Year').count()
CU_Haryana.groupby('Publication Year').count()
CU_Himachal.groupby('Publication Year').count()
CU_Jammu.groupby('Publication Year').count()
CU_Jharkhand.groupby('Publication Year').count()
CU_Karnataka.groupby('Publication Year').count()
CU_Kashmir.groupby('Publication Year').count()
CU_Kerala.groupby('Publication Year').count()
CU_Punjab.groupby('Publication Year').count()
CU_Rajasthan.groupby('Publication Year').count()
CU_SBihaar.groupby('Publication Year').count()
CU_TamilNadu.groupby('Publication Year').count()
HariSinghGour_University.groupby('Publication Year').count()
GuruGhasidas_University.groupby('Publication Year').count()
Hemawati_University.groupby('Publication Year').count()
JMI.groupby('Publication Year').count()
JNU.groupby('Publication Year').count()
Manipur_University.groupby('Publication Year').count()
Mizoram_University.groupby('Publication Year').count()
Nagaland_University.groupby('Publication Year').count()
NorthEasternHillUniversity.groupby('Publication Year').count()
Pondicherry_University.groupby('Publication Year').count()
RajivGandhi_University.groupby('Publication Year').count()
Sikkim_University.groupby('Publication Year').count()
Tezpur_University.groupby('Publication Year').count()
EngNForLan_University.groupby('Publication Year').count()
IndiraGandhiTribal_University.groupby('Publication Year').count()
Tripura_University.groupby('Publication Year').count()
AU.groupby('Publication Year').count()
DU.groupby('Publication Year').count()
HU.groupby('Publication Year').count()
ViswaBharti_University.groupby('Publication Year').count()

```

Further analysis like organisation and calculation of CAGR has been carried out using Google Sheets.

Code for making line plot of the publication count of central universities from 2010 to 2023

```

import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("data.csv")

# Plotting the line chart
plt.figure(figsize=(10, 6))
plt.plot(df['Publication Years'], df['Record Count'], marker='o', label='Record Count')

# Adding point labels
for x, y in zip(df['Publication Years'], df['Record Count']):
    plt.text(x, y, f'{y}', fontsize=9, ha='center', va='bottom')

# Chart labels and formatting
plt.xlabel('Publication Years', fontsize=12)
plt.ylabel('Record Count', fontsize=12)
plt.title('Record Count per Year', fontsize=14)
plt.xticks(df['Publication Years'], rotation=45)
plt.grid(True, linestyle='--', alpha=0.7)
plt.legend()
plt.tight_layout()

# Show the plot
plt.show()

```

B. Analysis of Thematic Trajectories of Research in the seven (07) Computer Science Categories of Web of Science

Code for plotting of publication count of different time periods

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df1 = pd.read_csv("ppdf_99_03.csv")
df2 = pd.read_csv("ppdf_04_08.csv")
df3 = pd.read_csv("ppdf_09_13.csv")
df4 = pd.read_csv("ppdf_14_18.csv")
df5 = pd.read_csv("ppdf_19_23.csv")
# Combine DataFrames into a list
datasets = [
    df1,
    df2,
    df3,
    df4,
    df5
]

# List of periods corresponding to each dataset
periods = ["1999-2003", "2004-2008", "2009-2013", "2014-2018", "2019-2023"]

```



```

# Loop through each dataset and create a line plot
for i, df in enumerate(datasets):
    # Create a complete range of years for the given period
    start_year, end_year = map(int, periods[i].split("-"))
    all_years = pd.Series(range(start_year, end_year + 1), name="Publication Year")

    # Count papers for each year and reindex to fill missing years with 0
    papers_per_year = df.groupby("Publication Year").size().reindex(all_years, fill_value=0)

    # Plot the trend as a line plot
    plt.figure(figsize=(8, 5))
    plt.plot(
        papers_per_year.index,
        papers_per_year.values,
        marker="o",
        linestyle="-",
        color="blue",
        label="Number of Papers",
    )
    plt.title(f"Trend of Research Papers Published ({periods[i]})", fontsize=14)
    plt.xlabel("Publication Year", fontsize=12)
    plt.ylabel("Number of Papers", fontsize=12)
    plt.xticks(papers_per_year.index, rotation=45, fontsize=10)
    plt.yticks(fontsize=10)
    plt.grid(axis="both", linestyle="--", alpha=0.7)
    plt.legend(fontsize=10)
    plt.tight_layout()

    # Show the plot
    plt.show()

# Show the combined plot for 2010-2023
combined_df = pd.concat(datasets)

# Create a complete range of years (1999 to 2023)
all_years = pd.Series(range(1999, 2024), name="Publication Year")

# Group by Publication Year and count papers, reindex to fill missing years with 0
papers_per_year = combined_df.groupby("Publication Year").size().reindex(all_years, fill_value=0)

# Plot the trend as a line plot
plt.figure(figsize=(10, 6))
plt.plot(
    papers_per_year.index,
    papers_per_year.values,
    marker="o",
    linestyle="-",
    color="green",
    label="Number of Papers",
)
plt.title("Trend of Research Papers Published (1999-2023)", fontsize=16)
plt.xlabel("Publication Year", fontsize=14)
plt.ylabel("Number of Papers", fontsize=14)
plt.xticks(papers_per_year.index[::2], rotation=45, fontsize=12) # Show every second year
plt.yticks(fontsize=12)

```

```
plt.grid(axis="both", linestyle="--", alpha=0.7)
plt.legend(fontsize=12)
plt.tight_layout()
```

```
# Save the plot
plt.savefig("papers_trend_1999_2023.png")
```

```
# Show the plot
plt.show()
```

Code for Word/Tag Cloud

```
!pip install wordcloud
```

```
from wordcloud import WordCloud, STOPWORDS
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df1 = pd.read_csv("ppdf_99_03.csv")
df2 = pd.read_csv("ppdf_04_08.csv")
df3 = pd.read_csv("ppdf_09_13.csv")
df4 = pd.read_csv("ppdf_14_18.csv")
df5 = pd.read_csv("ppdf_19_23.csv")
```

```
# List of dataframes
```

```
dataframes = [df1, df2, df3, df4, df5]
```

```
# Function to process data and generate word cloud
```

```
def process_and_generate_wordcloud(dataframes, column_name="Author Keywords"):
```

```
    for i, df in enumerate(dataframes):
```

```
        # Remove rows with null values in the specified column
```

```
        df = df.dropna(subset=[column_name])
```

```
        # Combine all keywords into a single string
```

```
        text = " ".join(df[column_name])
```

```
        # Generate word cloud
```

```
        wordcloud = WordCloud(width=800, height=400, background_color='white',
                                colormap='viridis').generate(text)
```

```
        # Display the word cloud
```

```
        plt.figure(figsize=(10, 5))
```

```
        plt.imshow(wordcloud, interpolation='bilinear')
```

```
        plt.axis("off")
```

```
        plt.title(f"Word Cloud for Period {i + 1}", fontsize=16)
```

```
        plt.show()
```

```
# Process and generate word clouds for the dataframes
```

```
process_and_generate_wordcloud(dataframes)
```

Code for Clustering

```

!pip install pandas scikit-learn matplotlib nltk
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import nltk
from nltk.corpus import stopwords
df = pd.read_csv("ppdf_99_03.csv")
df
df1 = df[['DOI', 'Author Keywords']]
df1
df1.dropna(subset=['Author Keywords'], inplace=True)
df1
# Preprocess the data
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))
# Function to preprocess text
def preprocess_text(text):
    # Remove semicolons and split into words
    words = text.replace('; ', ' ').split()
    # Remove stop words
    words = [word for word in words if word.lower() not in stop_words]
    return ' '.join(words)
# Apply preprocessing
df1['Processed Keywords'] = df1['Author Keywords'].apply(preprocess_text)
# Vectorization using TF-IDF
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(df1['Processed Keywords'])
# Elbow Method
wcss = []
for i in range(1, 12): # Try up to 11 clusters
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(10, 6))
plt.plot(range(1, 12), wcss)
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()

best_n_clusters = #optimal number of clusters where the elbow point in the graph is observed

# Clustering using K-Means with the optimal number of clusters
kmeans = KMeans(n_clusters=best_n_clusters, random_state=42)
kmeans.fit(X)
# Add cluster labels to DataFrame
df1['Cluster'] = kmeans.labels_

from wordcloud import WordCloud

# Generate word clouds for each cluster

```

```

for cluster in df1['Cluster'].unique():
    # Combine all keywords from the cluster
    text = ''.join(df1[df1['Cluster'] == cluster]['Author Keywords'])

    # Create a word cloud
    wordcloud = WordCloud(width=800, height=400, background_color='white').generate(text)

    # Display the word cloud
    plt.figure(figsize=(10, 5))
    plt.imshow(wordcloud, interpolation='bilinear')
    plt.axis('off') # Hide axes
    plt.title(f'Word Cloud for Cluster {cluster}')
    plt.show()

```

Code for Alluvial/Sankey Diagram

```

import pandas as pd
import plotly.express as px

# Load and preprocess dataset
data = pd.read_csv("ppdf_99_03.csv")
data.head()

# Create a function to map years into intervals
def get_time_period(year):
    if year == 1999:
        return "1999-2000"
    elif year == 2000:
        return "2000-2001"
    elif year == 2001:
        return "2001-2002"
    elif year == 2002:
        return "2002-2003"
    elif year == 2003:
        return "2003-End"
    else:
        return None # Handle edge case safely

# Map publication years to their respective time periods
data['Time Period'] = data['Publication Year'].apply(get_time_period)

# Drop rows without a valid time period (if any edge case exists)
data = data.dropna(subset=['Time Period'])

# Split keywords into individual items
data['Keywords'] = data['Author Keywords'].str.split(';') # Adjust separator if needed

# Explode the keywords into individual rows
data_exploded = data.explode('Keywords')

#convert to lower case
data_exploded['Keywords'] = data_exploded['Keywords'].apply(lambda x: x.lower() if isinstance(x, str) else x)

```

```

# Remove leading and trailing spaces from a specific column
data_exploded["Keywords"] = data_exploded["Keywords"].apply(lambda x: x.strip() if isinstance(x, str) else x)

# Group by time periods and keywords to calculate their frequency
keyword_counts = data_exploded.groupby(["Time Period", "Keywords"]).size().reset_index(name='Frequency')

# Sort the data by time periods
print(keyword_counts.head())

# Sum keyword counts across all time periods to find the top 30 keywords
top_keywords = keyword_counts.groupby('Keywords')['Frequency'].sum().nlargest(30).index
top_keywords

# Filter the dataset to include only these top 30 keywords
filtered_data = keyword_counts[keyword_counts['Keywords'].isin(top_keywords)]
import plotly.express as px

# Visualize only the top 30 keywords' evolution
fig = px.sunburst(
    filtered_data,
    path=["Time Period", "Keywords"],
    values="Frequency",
    color="Frequency",
)

fig.update_layout(
    title_text="Evolution of Top 30 Author Keywords in Computer Science (1999-2003)",
    margin=dict(t=50, l=25, r=25, b=25)
)

fig.write_html('/content/figure/alluvialplot9903.html') #to save an interactive plot

fig.show()

#sankey diagram with some improvements
import plotly.graph_objects as go

# Prepare nodes (time periods and keywords)
time_periods = filtered_data["Time Period"].unique()
keywords_list = list(top_keywords) # List of top 30 keywords

# Create a combined list of unique nodes for Sankey
all_nodes = list(time_periods) + keywords_list

# Map nodes to indices
node_indices = {node: idx for idx, node in enumerate(all_nodes)}

# Set up source, target, and values for Sankey diagram
source_indices = []
target_indices = []
values = []

for index, row in filtered_data.iterrows():

```

```

# Map time periods and keywords into indices
source_idx = node_indices[row['Time Period']]
target_idx = node_indices[row['Keywords']]

# Append to source/target lists only if frequency is non-zero
if row['Frequency'] > 0:
    source_indices.append(source_idx)
    target_indices.append(target_idx)
    values.append(row['Frequency'])

# Create Sankey diagram using Plotly
fig = go.Figure(go.Sankey(
    node=dict(
        pad=15,
        thickness=20,
        line=dict(color="black", width=0.5),
        label=all_nodes
    ),
    link=dict(
        source=source_indices,
        target=target_indices,
        value=values
    )
))

fig.update_layout(
    title_text="Transitions of Top 30 Keywords Over Time Periods (1999-2003)",
    font_size=12
)
fig.to_html('/content/figure/sankeyplot9903.html')
fig.show()

```